

Calorie Estimation Using Deep Learning For Dietary Images

A Capstone Project report submitted
in partial fulfillment of requirement for the award of degree

BACHELOR OF TECHNOLOGY

in

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE

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ACKNOWLEDGEMENT

We owe an enormous debt of gratitude to our Capstone project guide **Mr. Nawaz Pasha, Asst.Prof./Assoc Prof/Professor** as well as Head of the School of CS&AI , **Dr. M.Sheshikala, Professor** and Dean of the School of CS&AI, **Dr.Indrajeet Gupta Professor** for guiding us from the beginning through the end of the Capstone Project with their intellectual advices and insightful suggestions. We truly value their consistent feedback on our progress, which was always constructive and encouraging and ultimately drove us to the right direction.

We express our thanks to project co-ordinators **Mr. Sallauddin Md, Asst. Prof., and R.Ashok Asst. Prof.** for their encouragement and support.

Finally, we express our thanks to all the teaching and non-teaching staff of the department for their suggestions and timely support.

ABSTRACT

Studies around the world have shown that diet is one of the essentials for a healthy life. Everyone now a day is bothered about their diet more than ever. At present, due to its intrinsic association with almost all common chronic diseases including diabetes, heart stroke, renal disease and often cancer [1-3], obesity is regarded as one of the most important health problems in the world. The best prevention for obesity is the management of food intake which requires knowledge of food consumption like knowing calories in every bitstep. Fruits and vegetables are very essential in our everyday diet because they are the main source of energy. Fruits and veggies are packed full of vitamins and nutrients that can assist us in dealing with various health issues such as cancer, diabetes, heart ailments etc. Utilization of recorded data with the modern Day science and Technology it is possible to create a unique Identification system people can follow up their daily diet using logarithm as fruit and vegetable calorie intake system gets substantially effective for preventing ailments without expert dietitian advice. In a view to overcome these problems, this paper proposed an effective and robust deep learning model that not only results in high accuracy in classifying and predicting calories present in fruits and vegetables but also ensure user the flexibility of selection of image either appetizing or directly nutritious contents. So, to assess the intake of food and making them fit with sound quality for a lengthy life time we are providing dietary plans of various health issues like hypertension, rheumatoid arthritis, high-cholesterol etc.

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LIST OF ACRONYMS

Acronym	Abbreviation
ANN	Artificial Neural Network
BFC	Bayesian Fuzzy Clustering
CDCP	Centers for diseases control and prevention
CNN	Convolutional Neural Network
GUI	Graphical User Interface
IT	Information Technology
KNN	k-Nearest Neighbors
NIH	National Institutes of Health
R-CNN	Regions with Convolutional Neural Networks
RBF	Radial Basis Function
RBM	Restricted Boltzmann Machines
RELU	Rectified Linear Activation Unit
RF	Random Forest
RIS	Reverse Image Search
SIFT	Scale-Invariant Feature Transform
SSD	Single Shot MultiBox Detector
SVM	Support Vector Machine
WLM-NN	Whale Levenberg Marquardt Neural Network
WOA	Whale Optimization Algorithm

CHAPTER 1

INTRODUCTION

Nutrition is an essential foundation for the health and development of human life from the earliest stage of fetal development to old age. Healthy food and proper nutrition are undoubted requirements for survival, physical development, mental growth, performance and productivity, health and well-being [1-5]. The concept of nutritional informatics was created by combining information technology (IT), information science and nutrition. This branch of health informatics has become a new approach for registered dietitians and dietetic technicians to practice in the field and generate profit for the healthcare industry.

The first scientific use of the term "nutritional informatics" dates back to 1996; however, nutritionists have been applying nutritional information and anticipating technology for decades. The first paper showing the potential use of a computer to analyze dietary intake was published in 1962. Health care incorporated IT to implement patient care through electronic health records, collect population data through the collection of personal health records, and support clinical research through data reuse. . In addition, dietitians have incorporated specific electronic tools developed to manage patient tray service, indexing, and nutrient assessment [6–10] .

According to the NIH, obesity is the second leading cause of preventable death. Obesity is becoming a major health problem in many parts of the world; high calorie intake can be harmful and lead to obesity. Thanks to Western lifestyles, the Internet and television commercials, young people, including children, are consuming fast food and other junk food more often than before. It causes health problems such as heart disease, diabetes, blood pressure and cancer. A million people die every year due to obesity. Nowadays it is very difficult for a person to keep track of the calories he consumes. To deal with such problems, people pay more attention to what foods they consume. Previously, users used to track their calorie intake using spreadsheets. These methods are very hectic and lead to an unquantified diet. Food that is quantified will help you reduce excess fat. Calorie intake plays a very important role in a healthy lifestyle. Food is the key to the human body. So the diet plan must always take into account that you are aware of the total number of calories you need to consume in a day to maintain a healthy and fit life. But in

most cases, unfortunately, people face difficulties in estimating calories and making diet plans for sick people.

1.1 OVERVIEW

Health is one of the foremost important aspects of a person's life. It takes some amount of effort from an individual to remain in shape and maintain a healthy diet. Nowadays it's very difficult for someone to trace the calories consumed by them. Obesity becoming a significant health concern in many parts of the world, high calorie food intake is often harmful and leads to obesity. As per medical research, 1.9 billion people aged above 18 years are suffering from being overweight. The recent cases of obesity among the young generation are very high because the youth including children are more likely to consume nutrition and other junk foods than before. It causes the health problems like heart diseases, diabetes, blood pressure, and cancer in order to avoid such problems people are paying more attention to what form of food they're consuming. The intake of calories plays a vital role in one's healthy lifestyle. To avoid this issue each and every individual needs to opt for diet charts or plans to maintain their weight. In each and every diet plan fruits and vegetables are major sources of energy, vitamins, fiber, plant chemicals, and nutrients. These contribute to an essential part of our diet. A diet high in fruit and vegetables can help us against cancer, diabetes, heart diseases, etc. Along with health benefits, eating fruits can make weight management easier, according to the centers for disease control and prevention (CDCP). This paper proposes a web-based application that is used to classify, predict the calories in fruits-vegetable and also suggest dietary plans for different chronic diseases like high blood pressure, rheumatoid arthritis, high cholesterol, etc. Our system will automatically classify the image by using mobileNetV2 CNN architecture and it will give you predictions about the name of fruit/vegetable and its calorie, total fat count. It also provides weekly diet plans for diseases like high blood pressure, rheumatoid arthritis, high cholesterol, etc.

1.2 EXISTING METHODS

There are some methods for dietary assessment that include self-reporting and manually recorded instruments. However, it has some problems with estimating calorie consumption. participants' caloric intake is prone to bias, i.e. underestimating and underestimating food

intake to increase accuracy and reduce bias, current methods are enhanced by a mobile cloud computing system that uses devices such as smartphones to capture food intake. and calorie information. In addition to this step, the diet and calorie information is analyzed using cloud computing power for evaluation. However, users still have to enter information manually. Although much R&D effort has been devoted to the visual analysis of dietary and calorie information, efficient information extraction from food images remains a challenging problem. The Discrete Cosine Transform is used to portion food and is sought by the segmentation process. K-NN (k nearest neighbor) is used for classification, which takes a lot of time to train images and classify them. If data is not received correctly, data loss may occur.

1.2.1 Image Classification Techniques

1.2.1.1 K-Nearest Neighbors (KNN)

KNN is a simple image classification technique that tries to find a predefined number of training samples, the k-nearest neighbours that have the closest distance to a new sample. The first step of KNN is to memorize all the images data and their labels. Then, a distance function is used to measure distance between two images. The common distance function is L1 and L2 distance which are also known as Manhattan and Euclidean distance respectively as shown in Equation 1 and 2 below

$$(l1, l2) = \sum |l1 p - l2 p p| (1)$$

$$(l1, l2) = \sqrt{\sum (l1 p - l2 p)^2} (2)$$

The distance between each input samples are compared to every training sample and k number of training samples with closest distance to input samples is obtained and followed by their labels. A majority voting is performed on the predictions to get final predicted label for each input sample. In this case, the hyperparameter is value of k as it determines how many predictions to be considered in the voting. A majority voting is performed on the predictions to get final predicted label for each input sample. In this case, the hyperparameter is value of k as it determines how many predictions to be considered in the voting. An example of majority voting can be seen in Table 2-1 below. In this case, apple will be the final predicted label.

Table 1-1 Sample data to showcase majority voting

Labels	No. of Occurrences in predictions
Apple	5
Banana	2
Cabbage	3

The strength of KNN is its simplicity to implement as the algorithm is pretty straightforward while the weakness of KNN is its high computational cost as the size of the dataset grows larger. This will cause fast training as its just saving all data and slow prediction as it needs to compute distances between every training sample against every input samples.

Machine learning based food detection system

To develop an android-based implementation [12] for food recognition that could be used as a health awareness tool. The system allows user to take the photo of the food and show its nutritional contents. One goal of this study is also to determine the level of awareness of users in terms of their eating habits and food nutrition. Machine learning is used for detecting food from images. The study shows that, the respondent's level of food nutritional awareness is remarkably low as to the status of cognizance about food intake.

Automatic fruit classification

Hossam M. Zawbaa et al. [20] has presented machine learning techniques for automatic fruit classification. The classification is done using the Random Forest (RF) algorithm. The Random Forest (RF) algorithm result is compared with K-Nearest Neighborhood(K-NN) and Support Vector Machine (SVM) algorithms. For orange the height accuracy is achieved (100%) with SIFT feature extraction and KNN classifier. The height precision accomplished in the subsequent gathering is for 96.97%apple (with RF classifier), and for 78.89% orange (with SVM classifier).

1.2.1.2 Multiclass Support Vector Machine (SVM) Linear Classifier

To start with Multiclass SVM, it requires a loss function to tell how good the classifier is at predicting input dataset. The score function and loss function are as shown in Equation 3, 4 and 5 below. All the descriptions of symbols from the equations are in Table 2-2 as shown below.

$$f(x, W) = Wx \quad (3)$$

$$L = \frac{1}{N} \sum L_i(f(x_i), y_i) \quad (4)$$

$$L_i = \sum_{j \neq y} \max(0, s_j - s_{y_i} + 1) \quad (5)$$

Table 1-2 Symbols with Description for Multiclass SVM loss formulas

Symbol	Description
W	Weight
X	Input Samples
N	Number of samples
L_i	Loss for samples
S_j, S_{y_i}	Scores of true and predicted class computed from $f(x, W)$

In Equation 3, we compute the scores, s of each input samples. Then compare true class score against every predicted class score as seen in Equation 5. The true class score, s_y should be larger than predicted class score, s_j by margin of 1 to insure 0 loss, also known as hinge loss. Otherwise, set it to 0 as shown in Equation to prevent sum of negative loss. Lastly, simply divide by total number of samples to get overall loss as shown in Equation 4. To improve generalization and prevent overfitting, a regularization term is added to the computed loss usually. The most common examples are L1 and L2 regularization, the formulas are shown in Equation 6 and 7 below. L1 is just sum of weight while L2 is just sum of square of weight.

$$R(W) = \lambda \sum |w_i| \quad k \ i=1 \quad (6)$$

$$R(W) = \lambda \sum w_i^2 \quad k \ i=1 \quad (7)$$

Lastly, an optimization algorithm such as gradient descent, adam etc. is used to minimize the loss during the training session. The general gradient descent algorithm is shown in

Equation 8 below. Basically, it is just updating the weight by subtracting derivative term. A learning rate is used to control the speed of convergence towards local minima.

$$w(t+1) = w(t) - \alpha df(w(t)) dw \quad (8)$$

One of the strengths of multiclass SVM lies on its ability of generalization that creates simpler model. However, there are more hyperparameters to be considered and it is crucial to the performance of the classifier.

Use of SVM for classification

In the dietary management system [13], the calorie values are calculated by using segmentation, feature extraction and classification. After that, the calorie value is also calculated with food area volume and nutrition measure based on the mass value. A system [14] is proposed for identifying food item along with its nutrition elements. Bounding box are drawn around the food item in plate to identify it. Evaluation count of 81.55% is achieved.

Mobile cloud-based system

In the paper [15], authors proposed a structure for the measurement of calories. The cloud computing environment makes use of the Naive Bayes algorithm with machine learning for the purpose of food recognition. Images of food are clicked before and after eating. Food is also check for its freshness by using the textural extraction method. Three parameters are used for observing outcomes time, accuracy and freshness. Proposed system which is using Naive Bayes is more efficient as compared to systems which used SVM.

Augmented Reality based food recognition systems

It is a wearable gadget for food monitoring which is based on the concept of Augmented Reality. The system [11] allows to make the nutritional information of the food available and visible in various real –life scenarios, such as supermarket. Here, the use of reverse image search (RIS) and text mining has been done to recognize the food. The main components are food recognition, retrieval of nutrition information, image tracking and visualization. The overall accuracy of the RIS on images from real scene is 75.9% and on photos from the image database is 87.9%. The Google glass which is used for visualization run much hotter, also more power consumption and currently does not allow graphic processing unit acceleration on a regular Android smartphone and tablet. Therefore, the system cannot optimize the runtime much as things stand.

Nutrition monitoring systems based on wearable devices

A wearable gadget [17] based monitoring system for keeping track of nutrition intake. The data is provided to the wearable gadget through the use of handheld mobile device. Main task is to identify the gulping movements. A structure for assisting people with diabetes [18] which allows them to keep the track of their calorie count. Also, the goal is towards achieving daily food identification in semi-automatic manner. 75% of the classes showed improvement and the remaining 25% were mostly maintained.

1.2.1.3 DEEP LEARNING

Deep learning had gain huge success and applicable to wide area of applications such as voice recognition, image classification and games. It takes advantage of increasing computational power and availability of data nowadays. Deep learning actually inspired the way human brain works. Figure shows an artificial neuron and biological neuron which are very similar. Both of them work similarly by taking in inputs, process them and send out the output. Neuron receives multiple inputs and process it into one output. Each input is multiplied by a weight and add to a bias. During the training phase, the neural network adjusts the weights and biases accordingly through a process called backward propagation using gradient descent where calculations of gradient proceeds backwards from last layer to the first. This indicates that the neural network is learning from its data through minimizing the loss function.

There are many types of deep learning, Convolutional Neural Network is one of the most popular one. CNN is built up by a sequence of layers that transform an input to output with some differentiable function. The common layers in CNN are convolution layer, pooling layer and fully connected layer. First of all, Convolution layer is essentially a set of filters where each filter detects a feature. Its theory is similar to pooling where a sliding window scan across the inputs while each capture different features such as color, contours, edges etc. Fig.1.1 shows an example of the convolution layer. Then, another common layer is Fully Connected layer. In this layer, every neuron from one layer connects to the other layer. An illustration of Fully Connected layer is shown in Fig.1.2 below.

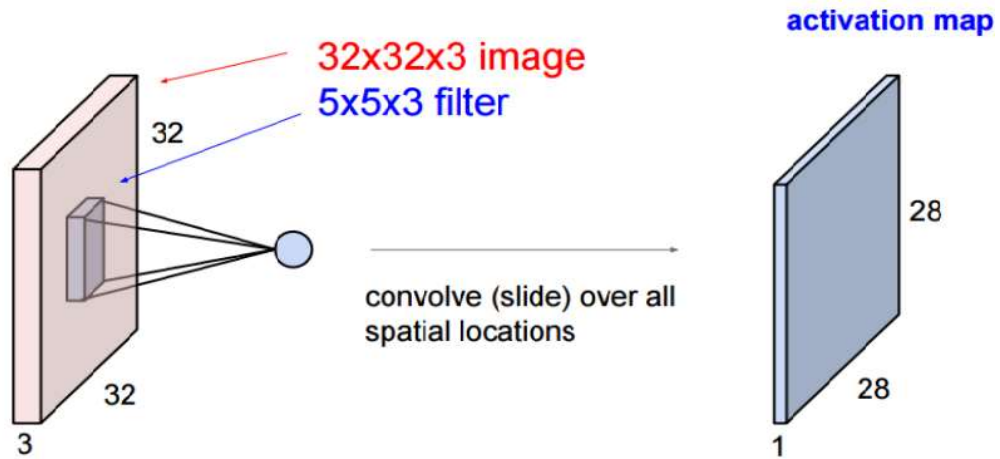


Fig.1.1: An example of convolution layer

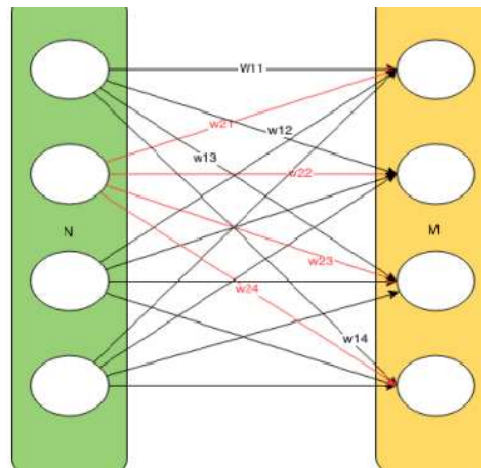


Fig.1.2: An example of fully connected layer

Besides that, sometimes overfitting can happen, this is the case where the network is too closely fit to the training set. It needs to be simpler and general in order to solve problems outside the training data. Therefore, dropout layer can help in this situation. This improves generalization as the model drops some of the neurons causing them to be deactivated.

1.3 PRESENT WORK

In the existing works only food calories are estimated for the particular food item in order to be aware of calorie intake without any specific diet recommendation to control food ingestion. Whereas in the present work, healthy weekly diet plans are suggested for the people who are suffering from diseases like Blood Pressure, Rheumatoid Arthritis, Cholesterol etc. along with estimation of calorie and fat content in fruits and vegetables.

1.3.1 OBJECTIVES

Healthy diet and nutrition are unquestionably necessary for survival, physical development, mental growth, performance and productivity. To maintain good health and for well-being of oneself the following objectives are being defined

- To propose a framework to estimate Calorie and fat content for different kinds of fruits and vegetables by applying deep learning.
- To identify the variety and healthy food for a diet.
- To quantify the intake of food for a diet based on diet plans.
- To make one fit with good health and longevity.
- To develop a functional prototype (web application) for the proposed framework.

1.3.2 PROPOSED MODEL

To develop a deep learning model which ensures both user flexibility and high accuracy in classifying and predicting the calories and fat content present in fruits and vegetables, as well as suggesting dietary plans for different chronic diseases like high blood pressure, rheumatoid arthritis, high cholesterol, and so on. In the present system fruit/vegetables are categorized and classified by utilizing mobileNetV2 CNN architecture and VGG16, which extracts features like color, shape, texture etc. for classification. After the fruit/vegetable has been classified, the calorie and fat content are retrieved or shown. Because we're using the MobileNetV2 architecture, the entire process is finished in milliseconds, resulting in quick responses with optimal accuracy.

1.4 LITERATURE SURVEY

Researches in the literature have often focused on different aspects of the food recognition problem. Many works address the challenges in the recognition of food by developing recognition strategies that differ in terms of features and classification methodologies. Here we briefly review some of the techniques like in [21], a system is proposed for the food recognition and calorie estimation using various models of machine learning. They used object detection to estimate the calories count of some famous Chinese dishes as well as that of Western dishes. Based on the food images and previously defined calorie data, they built some image-based calorie estimation models, which we hoped can accurately identify the name of the Chinese and Western foods and provide their calorie intake. The algorithms used are SSD (Single Shot MultiBox Detector) for real-time processing of object detection and classification. They also used “labelImg” a github

project to label the images. Software tools like jupyter notebook to execute codes.

W R Samemmanuel et al.[22] proposed a method which introduces the dietary assessment system through the efficient identification of the calorie values of each food items from the food images. The dataset contains various food or the meal images consumed by the person every day. The images in the food database are preprocessed for better processing. The proposed CSW-WLIFC algorithm does segmentation on each food. After segmentation, The feature extraction will extract features like shape, texture from the image. The proposed WLM-NN classifier performs to train the neural network with the existing LM model and the Whale Optimization Algorithm (WOA), which also performs classification of the food items from the image. Food items obtained from the WLM-NN classifier.

Hattarki Pooja et al. [23] proposed a system which takes as input an image of a food item and outputs the number of calories in this food item. The image is passed through MathWorks Image Processing Toolbox. The toolbox extracts visual features from the food item. The compressed image is passed through a classifier that predicts the type of food item in the image. The compressed image is also passed to a regressor which estimates the size of the image. Lastly, the compressed image is passed to another regressor that predicts the number of calories in the food.

Pallavi Kuhad et al. [24] proposed a method where the Category of tools is been considered, which uses image processing to recognize single and multiple mixed food objects, namely deep learning and SVM. Finger -based calorie measurement, distance measurement methods are applied for calorie estimation. A block resize method is also been proposed that uses the measured distance values along with the recognized food object name to further estimated calories.

S. Jasmine Miniya et al. [25] proposed a method where they pick a stable segmented region from multiple such segmented regions. After all images are segmented, the global and local features area is obtained via feature extraction based on texture, local neighborhood pixel and color. Once classification is performed, calorie value is obtained and an accuracy of 97% was obtained.

ParisaPouladzadeh et.al, [26] proposed calorie measuring system that illustrates health related problems faced by human being and to reduce such problems they introduced a system that can measure calories and nutrition in every day meals and this system can help patients and dieticians to measure and manage daily food intake. Via a special

calibration card technique, the algorithm gives accurate results.

Hong Liang et al. [27] proposed a method where the volume of food is estimated by 3-D reconstructions through depth camera or planar image sequence. Food calorie estimation is done using the Deep-learning approach. Datasets used are Microsoft COCO and Pascal VOC for image recognition. The calorimeter R2 and RMSE are about 0.95 and 43 and MSE is 32. Updating food ingredients information to meet the needs of different users. And gather larger training sets to improve accuracy and speed is aimed in the future.

ShotaSasano et al. [28] developed an auto food log record system with automatically estimate calories of food. The food recognition mainly consists of three steps: construction of food image database; extraction of effective visual features of food image such as color and texture features; image classification using machine learning methods such as Bayesian, Support Vector Machine and so on.

S. Jasmine Miniya et al. [29] proposed an automatic method of food category recognition and calorie estimation using the BFC and IpCA-DBN. For automatic recognition, the features are extracted, for which the optimal segments are offered using the BFC. The features increase the robustness of the recognition, and the method consumes less time, which is applicable for assessing the diet in daily life. Bayesian Fuzzy Clustering (BFC) is the optimal clustering mechanism, and the main role of BFC is to generate the optimal segments from the food image. BFC is characterized using a set of unknown memberships, and prototypes. DBNs are the generative neural networks that consist of the multiple layers of Restricted Boltzmann Machines (RBM) with each layer holding the input and the hidden neurons and the hidden neurons constituting the output layer. Once the food category is recognized using the proposed IpCA-DBN, the calorie of the food is determined.

GianluigiCiocca et al. [30] states that dietary assessment fundamental or backend plan is about dataset containing food classes and food images. Each image depicts a real canteen tray with dishes and foods arranged in different ways. Each tray contains multiple instances of food classes. The dataset contains 1027 canteen trays for a total of 3616 food instances belonging to 73 food classes. In this process they achieved an accuracy of 79% in calorie counting using CNN.

Gozde Ozsert Yigit et al. [31] proposed a method which uses Datasets Food11 and Food101. Deep convolution neural network structure is been trained from scratch by using different learning methods 1- Stochastic gradient descent, Nesterov's accelerated gradient,

and adaptive moment estimation. The tool used for pattern recognition problems is CNN, which is based on the cats' visual cortex method. The time taken for the training and testing process is very high. Adam and Nesterov's technique took almost 2.5 hours. Pre-trained food recognition structures with different classifiers are aimed in the future.

Chang Liu et al. [32] proposed to address these issues from the following two aspects: (1) to develop novel deep learning-based visual food recognition algorithms to achieve the best recognition accuracy; (2) to design a food recognition system employing edge computing-based service computing paradigm to overcome some inherent problems of traditional mobile cloud computing paradigm, such as unacceptable system latency and low battery life of mobile. In [33], another application-based system was proposed using feature-based artificial neural networks (ANNs) for the automatic recognition of Multilanguage handwritten digits. This is of particular interest in the fruit and vegetable calorie estimation system proposed in this paper because Multilanguage is a non-homogenous dataset similar to various fruits on a plate. In [34], the authors propose a system that uses CNN for health-related applications with the use of IoT to monitor and predict emergency health issues before they occur. The state-of-the-art system integrates many new technologies including cloud computing as well. Accuracy was not reported due to the lack of proper dataset.

In [35], the authors propose a technique based on SVM. The dataset consisted of pictures from Corel stock photos, and they were used as a classification to evaluate heavy tailed radial basis function (RBF) kernels, to surpass traditional polynomial or Gaussian RBF kernels in performance. The authors also discovered that a simple remapping can have an impact on the performance of the SVM's that run linearly to such an extent that, for such a problem, they become a valid RBF kernel alternative. Dense SIFT method and extracted visual vector using K-means clustering technique is being adopted in [36]. To support their work Support vector machine classifier was used to classify the food image and measures the carbohydrate level. calorie measuring system using special calibration card technique to measure calories and nutrition in every day meals to help patients and dieticians by measuring and manage daily food intake is presented in [37].

In [38] artificial neural networks (ANNs) for calorie estimation to various fruits on a plate using random forest algorithm along with structural features is presented. Epuru Sai Muralidhar et al. [39] proposes a faster R CNN-based food classification and food detection model using TensorFlow to estimate the number of calories. One of the Classification

Algorithms, a random forest algorithm is utilized to analyze and detect the data in the frame to decide the nourishment's form. An estimation strategy is made that approximates the quantity of calories from an image by enrolling the volume of the fruit or vegetable from the image and utilizing nourishment facts tables to figure out the calorie count in leafy foods. Manisha Mittal et al. [40] believes that this technique is nonexclusive in nature and they applied this model to various kinds of foods grown from the ground.

1.5 ORGANIZATION OF WORK

The complete project is organized into five chapters

Chapter 1: Describes the introduction to the fat and calorie intake and diet management, existing methods and proposed work, objectives and comprehensive literature survey.

Chapter 2: Describes the software and system requirements, libraries, algorithms required in implementing the work

Chapter 3: Presents the complete development and workflow part of our framework (model).

Chapter 4: Gives the results wherein the complete working of the model is described and analyzed.

Chapter 5: Presents the conclusions as well as the future scope of the work.

CHAPTER 2

HARDWARE AND SOFTWARE TOOLS

2.1 HARDWARE TOOLS REQUIRED FOR IMPLEMENTATION

The system requirements in developing our framework should have the following specification.

1. Windows Edition: Windows 10 Pro
2. Processor: Intel® Core™ [i5 processor] CPU
3. RAM: minimum 4.00 GB
4. System Type: 64-bit Operating System
5. HDD 10 GB Hard Disk Space and Above (Having SSD would be better)

2.2 SOFTWARE TOOLS REQUIRED FOR IMPLEMENTATION

2.2.1 PYTHON

It is an interpreted high-level general-purpose programming language. Python's design philosophy emphasizes code readability with its notable use of significant indentation. Its language constructs as well as its object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects. Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured (particularly, procedural), object-oriented and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library. Guido van Rossum began working on Python in the late 1980s, as a successor to the ABC programming language, and first released it in 1991 as Python 0.9.0. Python 2.0 was released in 2000 and introduced new features, such as list comprehensions and a garbage collection system using reference counting. Python 3.0 was released in 2008 and was a major revision of the language that is not completely backwards-compatible and much of Python 2 code does not run unmodified on Python 3. Python 2 was discontinued version 2.7.18 in 2020. Python consistently ranks as one of the most popular programming languages.

Technical Features

- Easy-to-learn – Python has few keywords, a simple structure, and a clearly defined syntax. This allows the student to pick up the language quickly.
- Easy-to-read – Python code is more clearly defined and visible to the eyes.
- Easy-to-maintain – Python's source code is fairly easy-to-maintain.

- A broad standard library – Python's bulk of the library is very portable and cross-platform compatible on UNIX, Windows, and Interactive Mode – Python has support for an interactive mode that allows interactive testing and debugging of snippets of code
- Portable – Python can run on a wide variety of hardware platforms and has the same interface on all platforms.
- Extendable – You can add low-level modules to the Python interpreter. These modules enable programmers to add to or customize their tools to be more efficient.
- Databases – Python provides interfaces to all major commercial databases.
- GUI Programming – Python supports GUI applications that can be created and ported to many system calls, libraries and windows systems, such as Windows MFC, Macintosh, and the X Window system of Unix.
- Scalable – Python provides a better structure and support for large programs than shell scripting

2.2.2 ANACONDA NAVIGATOR

Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda® distribution that allows you to launch applications and easily manage conda packages, environments, and channels without using command-line commands. Navigator can search for packages on Anaconda Cloud or in a local Anaconda Repository. It is available for Windows, macOS, and Linux.

To Getting started with Navigator section Fig.2.1 shows how to start Navigator from the shortcuts or from a terminal window. Navigator is an easy, point-and-click way to work with packages and environments without needing to type conda commands in a terminal window. You can use it to find the packages you want, install them in an environment, run the packages, and update them – all inside Navigator.

The following applications are available by default in Navigator:

- JupyterLab
- Jupyter Notebook
- Spyder
- PyCharm
- VSCode
- Glueviz

- Orange 3 App
- RStudio
- Anaconda Prompt (Windows only)
- Anaconda PowerShell (Windows only)
- Advanced conda users can also build their own Navigator applications.

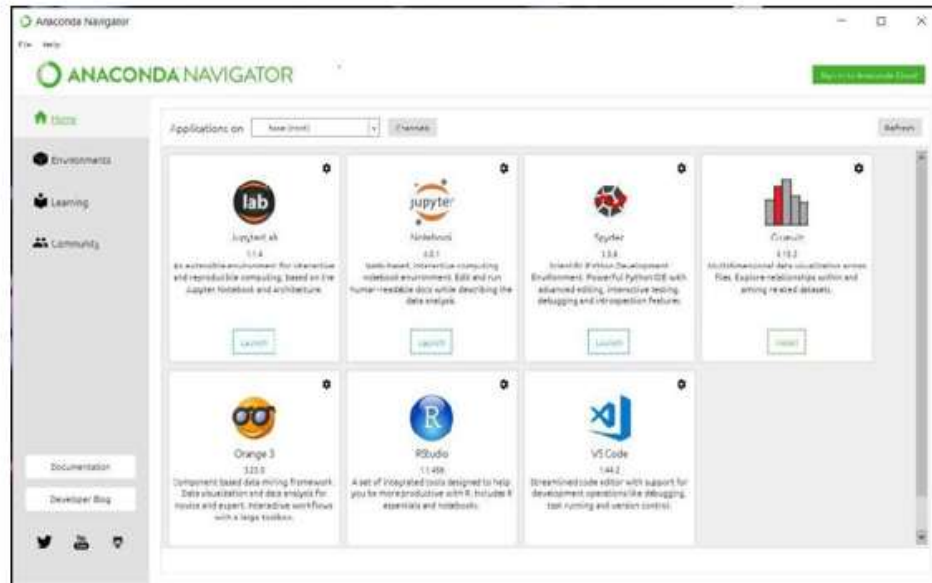


Fig.2.1:sample page of anaconda navigator

2.2.3 JUPYTER NOTEBOOK

Originally developed for data science applications written in Python, R, and Julia, Jupyter Notebook is useful in all kinds of ways for all kinds of projects:

1. Data visualizations
2. Code sharing
3. Live interactions with code
4. Documenting code samples

Most people have their first exposure to Jupyter Notebook by way of a data visualization, a shared notebook that includes a rendering of some data set as a graphic. Jupyter Notebook lets you author visualizations, but also share them and allow interactive changes to the shared code and data set.

Cloud services like GitHub and Pastebin provide ways to share code, but they're largely non-interactive. With a Jupyter Notebook, you can view code, execute it, and display the results directly in your web browser.

Jupyter Notebook code isn't static; it can be edited and re-run incrementally in real time, with feedback provided directly in the browser. Notebooks can also embed user controls (e.g., sliders or text input fields) that can be used as input sources for code.

If you have a piece of code and you want to explain line-by-line how it works, with live feedback all along the way, you could embed it in a Jupyter Notebook. Best of all, the code will remain fully functional—you can add interactivity along with the explanation, showing and telling at the same time.

2.2.4 TRANSFER LEARNING

Transfer learning is a machine learning method where knowledge learnt for a task is being reused on another task. In fact, very few people actually train an entire convolutional neural network, CNN from scratch. One of the reasons is that training a good network requires a huge number of datasets, time and resources. Instead, transfer learning had been a popular approach in deep learning where it could speed up training and improves performance. In other words, transfer learning is an optimization, it saves time and increase performance at the same time.

2.2.5 LIBRARIES

Python's large standard library, commonly cited as one of its greatest strengths provides tools suited to many tasks. For Internet-facing applications, many standard formats and protocols such as MIME and HTTP are supported. It includes modules for creating the graphical user interface, connecting to relational databases, generating pseudorandom numbers, arithmetic with arbitrary-precision decimals, manipulating regular expressions, and unit testing.

They are specified by their code, internal documentation, and test suites. However, because most of the standard library is cross-platform Python code, only a few modules need altering or rewriting for variant implementations. As of March 2021, contains over 290,000 packages with a wide range of functionality, including: Automation, Data analytics, Databases, Documentation, Graphical user interfaces, Image processing, Machine learning, Web frameworks, Web scraping.

TensorFlow:

TensorFlow, the open-source machine learning libraries is available for Python,

Java and C, it is a tool designed to build deep neural network models. Using TensorFlow, a built model can be deployed for usage on another platform including mobile platform, Android. This opened up wide opportunity for many new ideas to be implemented as it is impossible to perform machine learning in mobile platform before this without connecting to cloud. Besides, TensorFlow also supports deployment of computation across multiple CPU or GPU easily. This helps to save a lot of training time. Moreover, as compared to other machine learning library such as Theano, TensorFlow compile times is much better.

Keras

Keras is a high-level, deep learning API developed by Google for implementing neural networks. It is written in Python and is used to make the implementation of neural networks easy. It also supports multiple backend neural network computation. Keras is relatively easy to learn and work with because it provides a python frontend with a high level of abstraction while having the option of multiple back-ends for computation purposes. This makes Keras slower than other deep learning frameworks, but extremely beginner-friendly. Keras allows you to switch between different back ends. The frameworks supported by Keras are: TensorFlow, Theano, PlaidML, MXNet, CNTK (Microsoft Cognitive Toolkit).

Out of these five frameworks, TensorFlow has adopted Keras as its official high-level API. Keras is embedded in TensorFlow and can be used to perform deep learning fast as it provides inbuilt modules for all neural network computations. At the same time, computation involving tensors, computation graphs, sessions, etc can be custom made using the TensorFlow Core API, which gives you total flexibility and control over your application and lets you implement your ideas in a relatively short time.

Keras features

Keras prioritizes developer experience

- Keras is an API designed for human beings, not machines. Keras follows best practices for reducing cognitive load: it offers consistent & simple APIs, it minimizes the number of user actions required for common use cases, and it provides clear and actionable feedback upon user error.
- This makes Keras easy to learn and easy to use. As a Keras user, you are more productive, allowing you to try more ideas than your competition, faster -- which in turn helps you win machine learning competitions.
- This ease of use does not come at the cost of reduced flexibility: because Keras

integrates with lower-level deep learning languages (in particular TensorFlow), it enables you to implement anything you could have built in the base language. In particular, as `tf.keras`, the Keras API integrates seamlessly with your TensorFlow workflows.

NumPy

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays. It is the fundamental package for scientific computing with Python. Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using NumPy which allows NumPy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyse. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Pillow

Pillow is a fork of PIL that adds some user-friendly features. PIL is the Python Imaging Library. The Python Imaging Library adds image processing capabilities to your Python interpreter. This library provides extensive file format support, an efficient internal representation, and fairly powerful image processing capabilities. The core image library is designed for fast access to data stored in a few basic pixel formats. It should provide a solid foundation for a general image processing tool. The Pillow library contains all the basic image processing functionality. You can do image resizing, rotation and transformation. Pillow module allows you to pull some statistics data out of image using histogram method, which later can be used for statistical analysis and automatic contrast enhancement.

Beautiful Soup (HTML parser):

Beautiful Soup is a Python package for parsing HTML and XML documents (including having malformed markup, i.e., non-closed tags, so named after tag soup). It

creates a parse tree for parsed pages that can be used to extract data from HTML, which is useful for web scraping.

Streamlit

Streamlit is an open-source python framework for building web apps for Machine Learning and Data Science. We can instantly develop web apps and deploy them easily using Streamlit. Streamlit allows you to write an app the same way you write a python code. It is compatible with major Python libraries such as scikit-learn, Keras, PyTorch, SymPy(latex), NumPy, pandas, Matplotlib etc.

2.2.6 ALGORITHM USED

MobileNet

The current trend in deep neural networks is to go deeper and become more complicated. However, the improvements in accuracy does not make the network more efficient in terms of size and speed. In fact, real world applications require the recognition task to be performed in timely fashion on a computationally limited platform. For instance, of recent popular real-world applications are augmented reality, self-driving vehicles, robotics etc. Hence, this serves as a motivation for proposal of MobileNets.

The core idea of MobileNets is to build networks an efficient network with low latency and size which can be easily embedded into mobile or other embedding devices. In other words, model speed size is being heavily focused. First of all, MobileNets are based on a form of factorized convolution named depthwise separable convolution. Basically, a depthwise separable convolution factorizes a standard convolution into a depthwise convolution and a 1x1 convolution, also named pointwise convolution. The first layer is for filtering while the latter is for combining. Fig.2.3 illustrate the process of factorizing a standard convolution into a depthwise convolution and a pointwise convolution. As a result of this factorization, computation and model size had been drastically reduced.

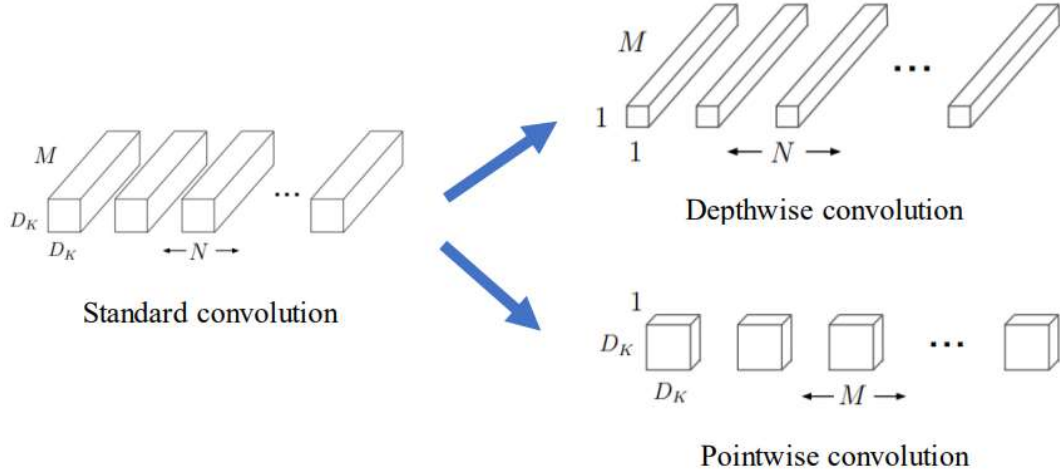


Fig.2.2: Standard convolution factorized into two separate layers named depthwise convolution and pointwise convolution

The idea of depthwise separable convolution is to reduce the computational cost comparing to a standard convolution. A comparison of computation cost for a standard convolution and a depthwise separable convolution is shown in Equation 9 and 10.

$$D_K * D_K * M * N * D_F * D_F \quad (9)$$

$$D_K * D_K * M * D_F * D_F + M * N * D_F * D_F \quad (10)$$

Table 2-1 Description of symbols for computation cost

Symbols	Description
D_K	Convolution kernel size
D_F	Spatial width & height of input
M	No. of channels of input
N	No. of channels of output

Depthwise separable convolution is used to break the interaction between size of kernel and amount of output channels. Depthwise convolutions are very computationally efficient compared to a standard convolution. However, a depthwise convolution only filter input channels and does not combine them to create new features. This is where pointwise convolution comes into play in order to generate these new features. As a result, MobileNet uses 3x3 depthwise which reduced computational cost by 8 to 9 times comparing to a standard convolution. With that being said, the drawback is that the accuracy is slightly worse than standard convolution. Beside the last fully connected layer, all layers in

MobileNet are accompanied with a batchnorm and ReLU nonlinearity. A comparison of standard convolution layer with a depthwise separable convolution is shown in Fig.2.4.

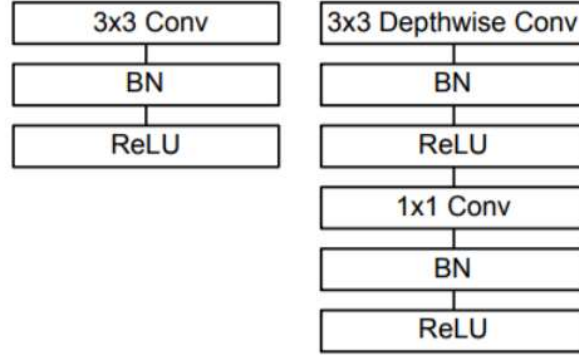


Fig.2.3: Standard convolutional layer followed by batchnorm and ReLU (left) vs depthwise separable convolution followed by batchnorm and ReLU

While the MobileNet architecture consisting of 28 layers by counting depthwise and pointwise convolutional layers as separated layers is already small and fast enough, it could still be made even smaller and faster. To achieve that, a parameter known as width multiplier, a is introduced. The sole purpose of this parameter is thinning a network at each layer uniformly. With that being said, the new computational cost of a depthwise separable convolution with width multiplier is as Equation 11.

$$D_K * D_K * aM * D_F * D_F + aM * aN * D_F * D_F \quad (11)$$

The baseline MobileNet has $a = 1$ while $a < 1$ are reduced MobileNets. According to Howard, the width multiplier can reduce the number of parameters quadratically and computational cost by roughly a^2 . Following width multiplier is another hyperparameter known as resolution multiplier, p . This multiplier is applied to input image thus change the computational cost again as shown in Equation 12.

$$D_K * D_K * aM * pD_F * pD_F + aM * aN * pD_F * pD_F \quad (12)$$

The same rules apply for resolution multiplier as $p = 1$ for baseline MobileNets and $p < 1$ for reduced MobileNets. As a result, resolution multiplier can reduce computational cost by p^2 . Because of the resolution multiplier is set implicitly, the input resolution of network is 128, 160, 192 and 224.

In the year of 2018, MobileNetV2 proposed as the successor to the original MobileNet as proposed in previous year. Same as the goal of its predecessor which is to introduce a new architecture that is specially designed for mobile and resource constrained environments. Sandler tries to decrease the number of operations and memory required

while retaining the same accuracy by introducing novel layer module, the inverted residual with linear bottleneck. This new module reduces the memory footprint during inference significantly in mobile environment.

First of all, MobileNetV2 retains the use of 3x3 depthwise separable convolution as in the first MobileNet. However, there are some changes to it. An additional 1x1 layer before the depthwise convolution while the pointwise layer now acts as projection or bottleneck layer. The newly added 1x1 layer will act the exactly opposite to the pointwise convolution which is to expand dimension before depthwise convolution. Secondly, residual connection was introduced to the same block which work just like those in ResNet. The existence of residual connections aids the flow of gradient during backward propagation. The modified version of the original depthwise separable convolution block is now named as residual bottleneck block. An example of a residual bottleneck block is shown in Fig.2.5 as compared to the original block.

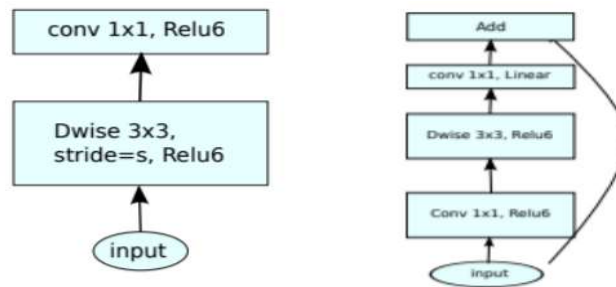


Fig.2.4: Comparison between original depthwise separable convolution block vs residual bottleneck block

The motivation behind these changes were due to the smaller the tensor, the fewer the multiplications the convolutional layers have to do. Thus, the lower the computational cost. Sandler et al. (2018) concluded with stating MobileNetV2 achieving better results than its predecessor in terms of accuracy and computation time and memory efficiency.

VGG16:

In our project, VGG16, a deep convolutional neural network pre-trained on ImageNet, is utilized as a feature extractor for food image classification. Its 16-layer architecture captures intricate visual patterns, enabling accurate identification of food items. By leveraging transfer learning, VGG16 provides a robust foundation, improving efficiency and accuracy in calorie estimation and diet recommendation tasks.

CHAPTER 3

PROJECT IMPLEMENTATION

3.1 SYSTEM DESIGN

The System architecture of the model implemented is shown in the Fig.3.1 here the calorie content present in fruits and vegetables is estimated by means of images collected from the dataset. Based on the type of disease a diet plan is recommend. In present work the diet plan is created for few diseases like blood pressure, cholesterol, diabetes, weight loss, PCOD, thyroid, rheumatoid arthritis.

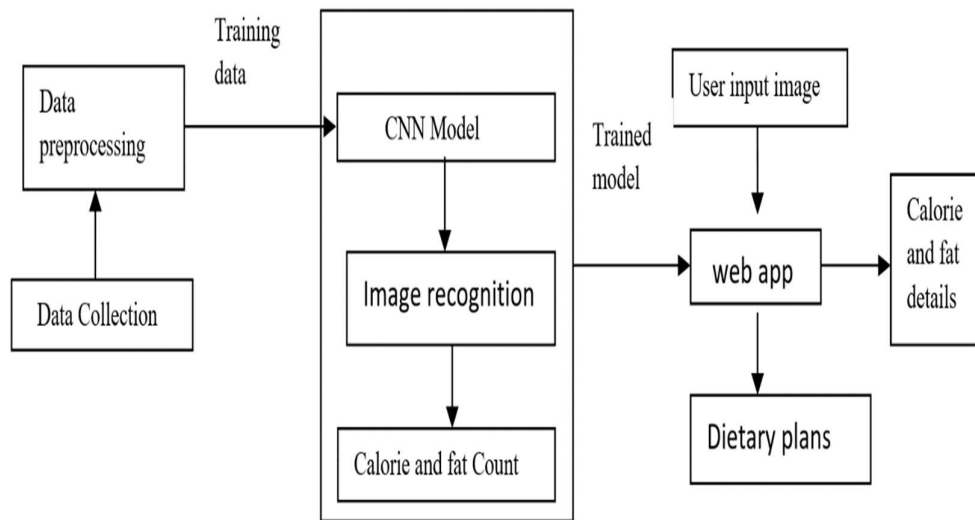


Fig.3.1: System architecture of proposed model

In order to obtain better results a deep learning model namely Convolution neural network (CNN) is used to improve the accuracy. In present work, a web application is designed to provide users with rapid and accurate results about the number of calories present in the image along with the fat content present in the fruits and vegetables that is consumed by an individual in their diet. Once, the image is uploaded and submitted in JPEG or PNG format, the uploaded image will first undergo pre-processing by resizing the image based on the model. After pre-processing, the image is optimized and sent to the next stage called classification, where CNN classification is done by using MobileNetV2 CNN model and a bit of VGG16. Here extraction of features like shape, colour, texture, etc. are done. Once the classification is done, the calorie and fat count of predicted fruit/vegetable are obtained. On the other hand, along with the calorie count and fat content the user can check

weekly diet plans for different diseases. The complete process is done within few milliseconds thereby provided fast answers with optimized accuracy. The various steps adopted in our system is explained in the following sections.

3.1.1 DATA COLLECTION

To implement this project, we are using the "Fruits and vegetable Image recognition" dataset. The data is divided into various class labels and the images are in .jpeg format. In the present dataset 36 different categories are taken which include Fruits- banana, apple, pear, grapes, orange, kiwi, watermelon, pineapple, mango. Vegetables- cucumber, carrot, onion, potato, lemon, tomato, radish, beetroot, cabbage, lettuce, cauliflower, chili pepper, turnip, sweet potato, paprika, jalapeno, ginger, peas, eggplant etc and in collectively there are 3600 training images and 10 images per each class in train and validation data.

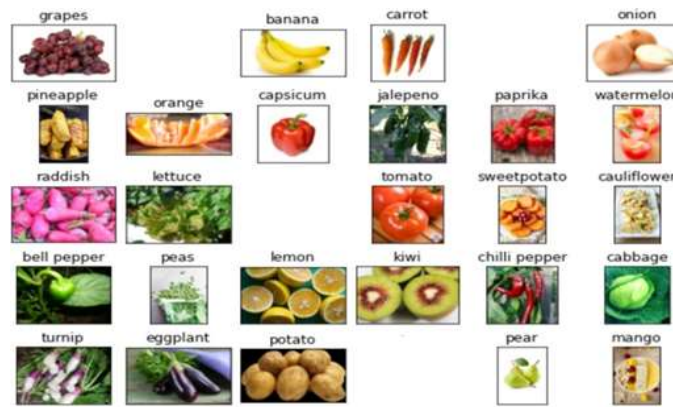


Fig.3.2: Sample dataset images

3.1.2 DATA PRE-PROCESSING

The second part of project was to process and prepare dataset required for later phase. In this project, images consisting of 36 kinds of fruits and vegetables including both local and western were collected from the internet. Then, these images were filtered out manually. The purpose of this step is to remove images that are irrelevant and it is crucial in order to achieve a good result. In order for the recognition to take place, the image has to be loaded into the system. This is done by choosing an RGB file from a directory. The image is then passed onto the pre-processing and resizing the images to 224x224 dimensions in order to speed up the performance of model. In order to increase the size of

the sample data taken data augmentation technique is used. In this case, keras ImageDataGenerator module which has an ability to generates new images from existing image automatically is used when the module trained. Generally, it does some techniques like Rotation, Width shift, Height Shift, Horizontal and vertical flip, Zooming, rotation and random brightness to generate new images for a single image and improve the network performance.

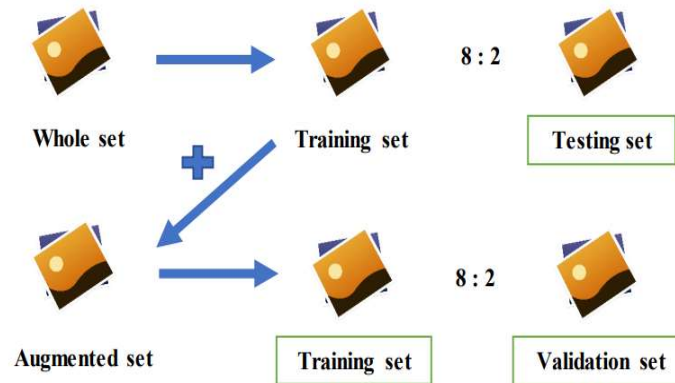


Fig.3.3: Splitting of dataset

As shown in Fig.3.2, the original image dataset consisting of 3825 images were split at 8:2 ratio as training and testing set respectively. Then, the augmented images were then appended to the training set followed by a split at 8:2 ratio as training and validation set. Finally, three datasets named training, testing and validation were prepared. On the other hand, all image labels are one-hot encoded as shown in Table 3.1. Basically, the process is to convert labels to binary format instead of integers to represent each class.

Let (apple, banana, carrot) = (0, 1, 2)

Table 3-1 Example results of one-hot encoding

Label(class)	One-hot encoding
Apple	(1,0,0)
Banana	(0,1,0)
Carrot	(0,0,1)

3.1.3 DATA MINING

Transfer learning using the weights from the pre-trained MobileNetV2 model was applied on the fruits and vegetable images to facilitate multi class classification of labels. TensorFlow was used to train the final model on the food dataset.

3.1.4 FINE TUNING MOBILENETV2

The next part of the project was to select an existing that is suitable for the use in the project case. Training sequential CNN model from scratch is time-consuming and resource intensive to define and train a good model. Therefore, approach of transfer learning was applied. However, there are many models available and a model selection is required. In order to select the best model for the project we considered model performance, size, and inference speed in devices. So, we MobileNetV2 is the best fit for the project. The weights and model architecture were downloaded from Keras API. Then, only few of the uppermost layers were being trained while the rest of the layers were all frozen. This helps as the earlier layers learnt simple and general features and the knowledge can be transferred to solve other problems.

Using MobileNetV2, the initial total trainable parameters is 3,504,872 as shown in Fig.3.4 as the MobileNetV2 was pre-trained on ImageNet dataset which consist of 1000

block_16_project (Conv2D)	(None, 7, 7, 320)	307200	block_16_depthwise_relu[0][0]
block_16_project_BN (BatchNorma	(None, 7, 7, 320)	1280	block_16_project[0][0]
Conv_1 (Conv2D)	(None, 7, 7, 1280)	409600	block_16_project_BN[0][0]
Conv_1_bn (BatchNormalization)	(None, 7, 7, 1280)	5120	Conv_1[0][0]
out_relu (ReLU)	(None, 7, 7, 1280)	0	Conv_1_bn[0][0]
global_average_pooling2d_1 (Glo	(None, 1280)	0	out_relu[0][0]
Logits (Dense)	(None, 1000)	1281000	global_average_pooling2d_1[0][0]
=====			
Total params: 3,538,984			
Trainable params: 3,504,872			
Non-trainable params: 34,112			

Fig.3.4: Last few layers of MobileNetV2 architecture before modification

Classes. The next step was to modify the last layer with the number of classes the project will have. In this case, the project will be covering 36 classes of desserts. The diversity of features learnt from ImageNet can be reused to solve different problems. In order to achieve that, the earlier layers of the architecture must be frozen and not be trained on new datasets. This is because the features learnt in earlier layers are general and not as specific as those

learnt from last few layers. Nonetheless, all the layers configurations were kept default. At this stage, the model should be ready to be trained on the new fruit and vegetable datasets that were prepared.

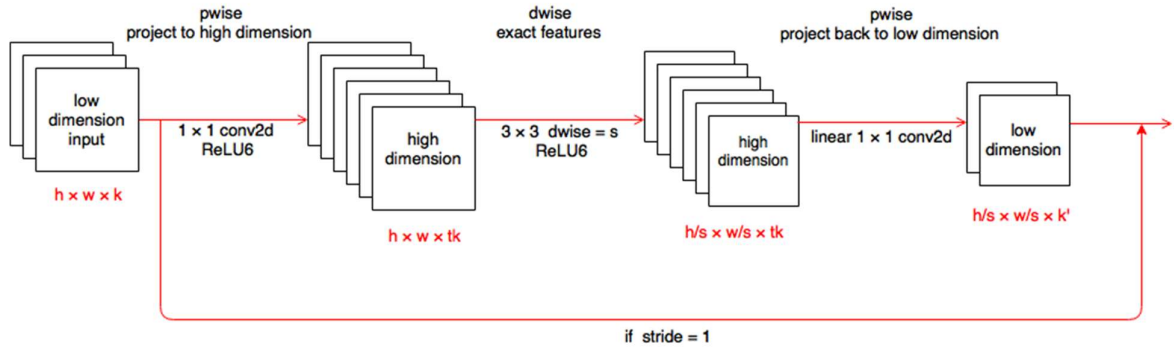


Fig.3.5: MobileNetV2 Architecture

3.1.5 TRAINING PHASE

The training phase is a time-consuming process and it requires heavy computing power. First of all, all images were resized into resolution of 224x224 with 3 color channels (RGB format) each. This is the default input resolution as recommended by MobileNetV2. Then, training and validation sets that were prepared previously were being used accordingly during the training session. In addition, there will be times where training loss started to increase and accuracy degrades after certain epochs of steady improvements. An example of this situation was shown in Fig 3.6. This can be solved by applying a technique called early stopping.

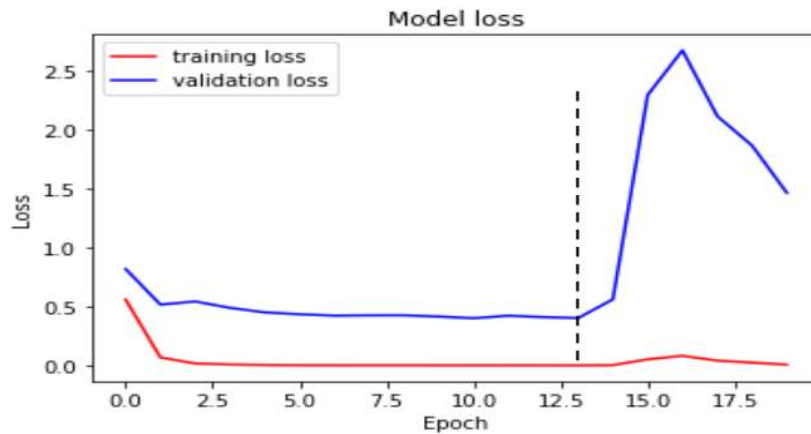


Fig.3.6: Model loss stopped improving after steady decrement

In our model a criterion is made to stop the training process once the training loss stopped improving such as stop training after no improvements were being seen after 2 epochs by providing patience as 2. This helps to secure the best model from training phase.

3.1.6 PERFORMANCE EVALUATION AND HYPERPARAMETER TUNING

After the training phase, the model accuracy was evaluated with testing set. The hyperparameters were being fine-tuned in order to optimize the performance. Some of them including learning rates, batch size etc. Accuracy and loss are the main metrics on which the implemented model is evaluated. The model was executed for 5 epochs on 36 classes of fruits and vegetables. So, with the execution of each epoch the accuracy and loss are recorded. The final model was executed for 5 epochs. The model had a training accuracy of 94.17% and validation accuracy of 95.1%. While the training loss and validation loss was recorded to be 0.1790 and 0.9521 respectively as shown in Fig.3.7.

```
100/100 [=====] - 226s 2s/step - loss: 1.9120 - accuracy: 0.4933 - val_loss: 0.4976 - val_accuracy: 0.8383
Epoch 2/5
100/100 [=====] - 189s 2s/step - loss: 0.7049 - accuracy: 0.7820 - val_loss: 0.2377 - val_accuracy: 0.9341
Epoch 3/5
100/100 [=====] - 175s 2s/step - loss: 0.4369 - accuracy: 0.8675 - val_loss: 0.1612 - val_accuracy: 0.9521
Epoch 4/5
100/100 [=====] - 182s 2s/step - loss: 0.2701 - accuracy: 0.9226 - val_loss: 0.1267 - val_accuracy: 0.9551
Epoch 5/5
100/100 [=====] - 184s 2s/step - loss: 0.1790 - accuracy: 0.9417 - val_loss: 0.1094 - val_accuracy: 0.9521
```

Fig.3.7: Loss and Accuracy on 5 epochs for 36 categories

3.1.7 MODEL DEPLOYMENT

After acquiring a satisfied model accuracy, the model architecture together with its weights were all frozen, saved as h5 file and converted into constants using TensorFlow library. Firstly, the load model library from the tensorflow.keras.models package is used to load the best model to access anytime. After saving the model, web app is developed using streamlit in local machine by installing required libraries as mentioned in software tools. The application code of streamlit consists of GUI for app, model processing code, image saving and prediction code and code for fetching the calories and total fat count for a particular class using Web scraping.

3.1.8 CALORIE EXTRACTION

Once the image is categorised and classified into fruit/vegetable the labels are mapped accordingly, and the class of input is predicted. The predicted class is sent as an

input to extract or retrieve calories from websites using web scraping. Here BeautifulSoup and requests are used for scrapping the calories and fat content from the internet for the predicted object.

3.1.9 RECOMMENDING WEEKLY DIET PLANS

The weekly dietary plans are suggested for the people who are suffering with diseases like high blood pressure, rheumatoid arthritis, high cholesterol etc. For this the user need to click on go to diet plan, at the bottom the user can see two options like “read more” and “view diet plan”. In the view diet plan the user can able to see weekly diet plan for respective disease and in the read more the user can able to view the information regarding disease like causes symptoms etc

3.2 FLOW INSIDE MOBILENETV2

Fig.3.9 is an example bottleneck residual block used in MobileNetV2. Basically, the block is built from several convolutional layers. Each projection (bottleneck) layer was followed by batch normalization while the rest of the convolution layers were all followed by batch normalization and ReLU non-linearity. The overall MobileNetV2 architecture is built from stacking and repetition of these blocks together with their residual connections as shown in Fig.3.10.

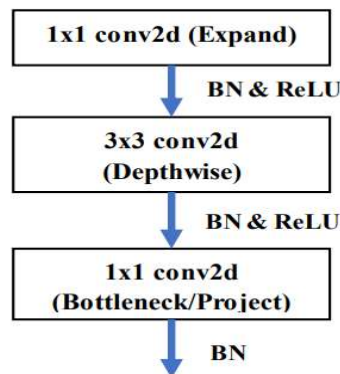


Fig.3.8: An example of bottleneck residual block in MobileNetV2

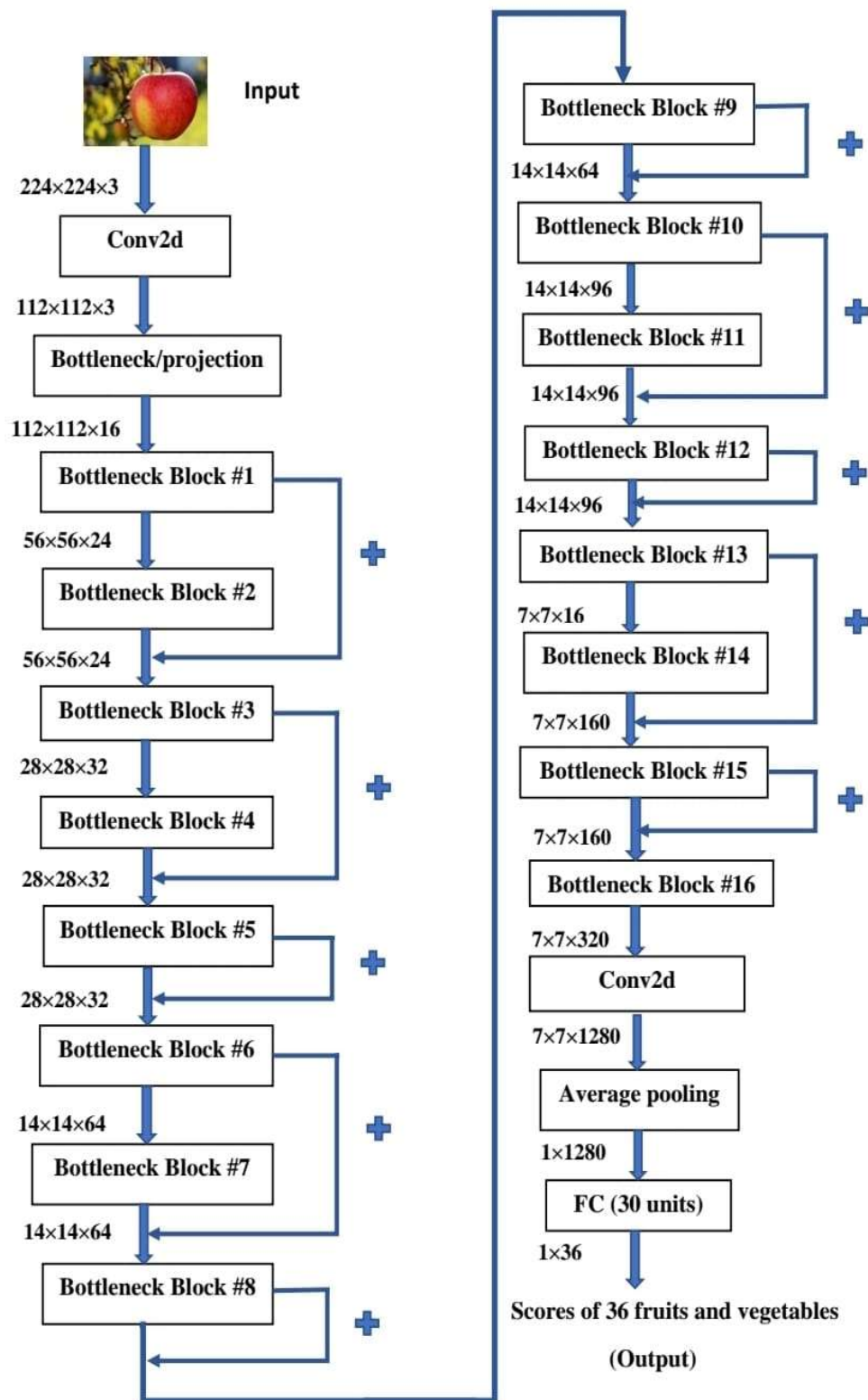


Fig.3.9: Flow of information inside MobileNetV2 showing the output dimension changes and residual connections going up the network

3.3 Workflow of Calorie and Fat content Extraction

For realization of the project, the flow of the project can be summarized into two parts as shown in Fig.3.10 and Fig.3.11.

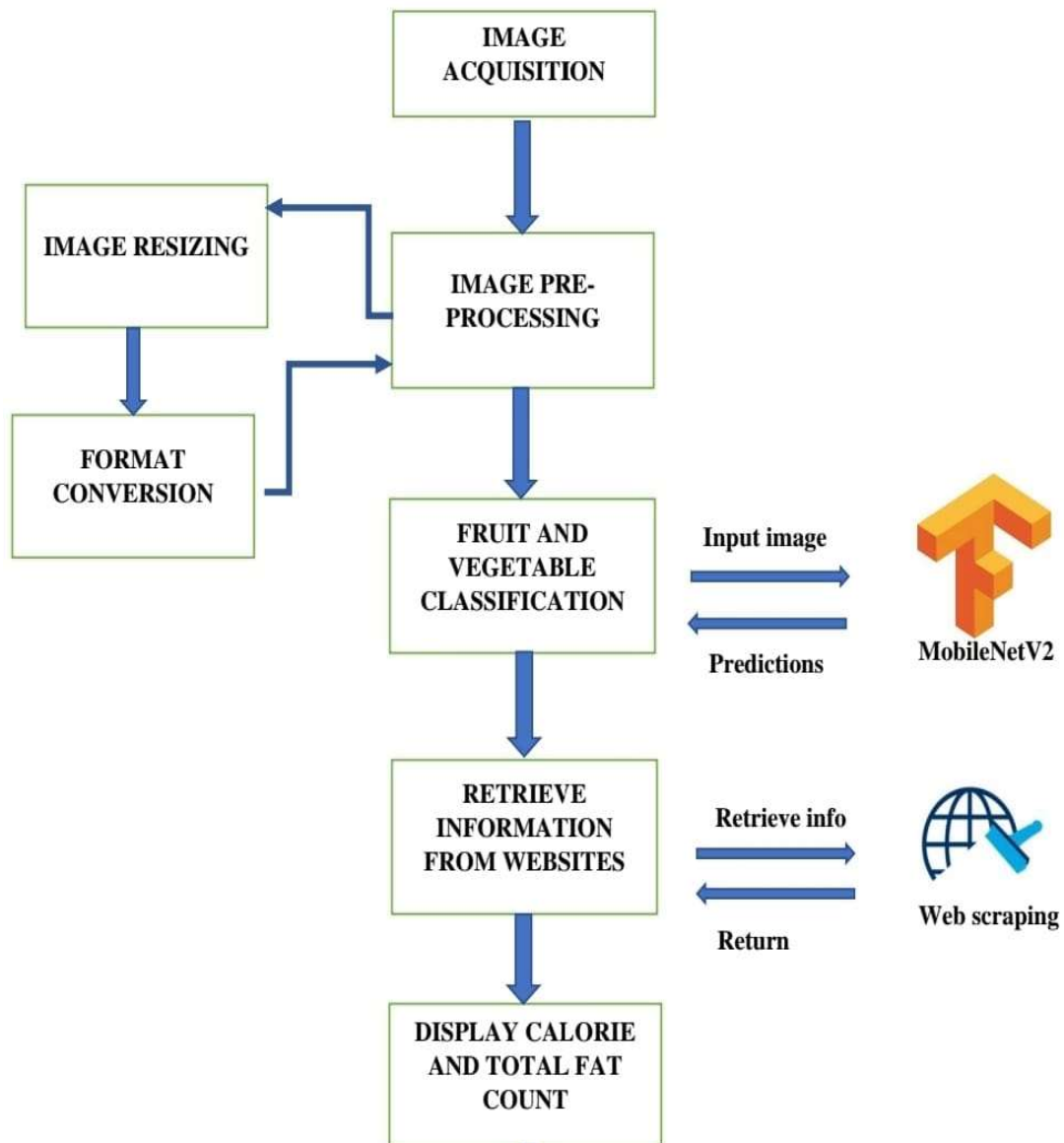


Fig.3.10:Detailed flow of calorie and fat content extraction

Step 1: Image Acquisition:

- The user will upload the image of fruit/vegetable, where the information of the data is collected from the dataset.
- The image that is uploaded is stored in the local System.

- The whole portion of the food should be present within frame without any blockage.
- Another way to upload a image is that, user can download the image of fruit/vegetable from the Google instead of collecting data from the dataset.

Step 2: Image pre-processing:

- Every time a new image is acquired, certain pre-process needs to be done before it is ready for interface.
- To start with that, each image is first resized into resolution of 224x224 according to the default set by MobileNetV2.
- Then, the array is arranged so that image channels are ordered last. The desired image format will be ordered by batch size, image width, image height, and number of color channels.
- Finally, the array is flattened into a two-dimensional array.

Step 3: fruit/vegetable Classification:

- The trained TensorFlow model will be used to perform fruit/vegetable classification.
- This is achieved through the use of TensorFlow interface. Interface API which acts as a bridge between TensorFlow model and the Web application.
- The pre-processed bitmap image is fed into the model for interface and the output predictions will be an array of scores.
- The scores are sorted and the fruits/vegetables class who scored the highest will be the final output.

Step 4: Retrieve information from websites:

- Since the fruits/vegetable classes are predicted by the model is ordered the same as the model output, the index of fruit/vegetable who scored highest will be used to query for its relevant information from websites using web scraping.

Step 5: Calorie and fat count display:

- The calorie and fat content present in the fruit/vegetable is obtained.
- For several serving Sizes, the calorie and fat content are computed and the final result is displayed to the user.

3.4 Workflow of Weekly Diet plan Recommender

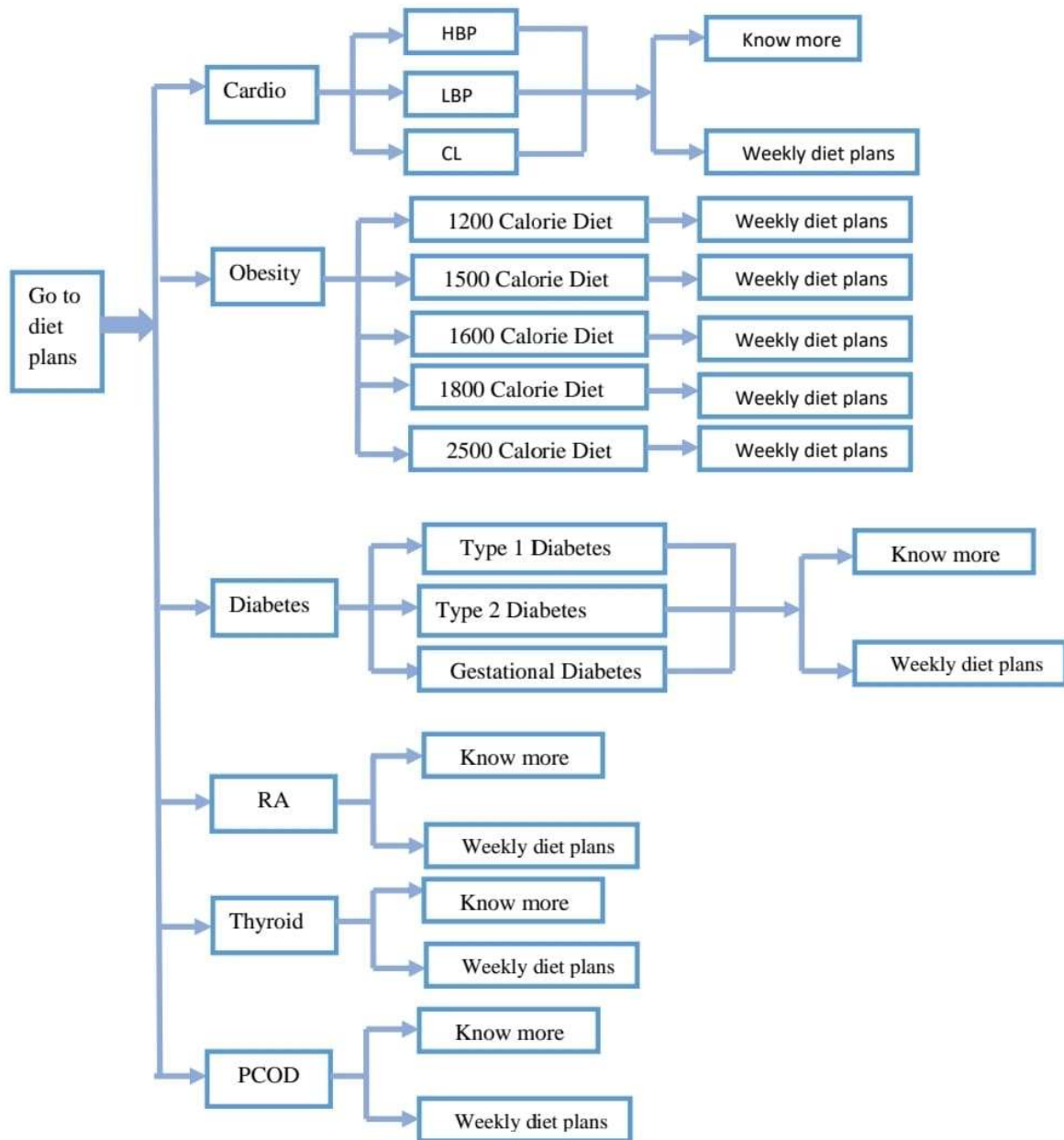


Fig.3.11: Detailed flow of weekly diet plan recommender

1. In this case, the dietary plans are recommended for the people who are suffering with some diseases like high blood pressure, rheumatoid arthritis, high cholesterol etc.
2. The user needs to click on the view diet plans so that they can able to view diseases like Cardio, Obesity, Diabetes, rheumatoid arthritis, Thyroid, Polycystic Ovarian Disease (PCOD).

3. In the cardio there are sub division called High blood pressure (HBP), Low blood pressure (LBP), Cholesterol. So, the user can able to view the information regarding the diseases as mentioned above along with that weekly diet plans are also suggested.
4. The user can view the weekly dietary plan for the obesity which has 1200, 1500, 1600, 1800, 2500 calorie diet.
5. Diabetes consists of Type1 diabetes, Type2 diabetes, and gestational diabetes, the information regarding the different diabetes and weekly dietary plan for each can be seen.
6. And the remaining diseases like rheumatoid arthritis, Thyroid, Polycystic Ovarian Disease (PCOD). We can view the information and weekly dietary plans.

CHAPTER 4

SIMULATION RESULT AND ANALYSIS

The trained model on 36 kinds of fruits and vegetables was tested against on some fruit and vegetable images to verify the model accuracy as shown in Fig.4.1. The trained model is deployed into a web application using python open-source framework Streamlit. So, the frontend and backend are being handled by the Streamlit. As a normal user, the user will visit our application by URL. On the website as shown in Fig.4.2 will have two icons one is to “browse files” to upload input image to the model and the other is to “go to diet plans” to view weekly diet plans for some diseases like high blood pressure, rheumatoid arthritis, high cholesterol.

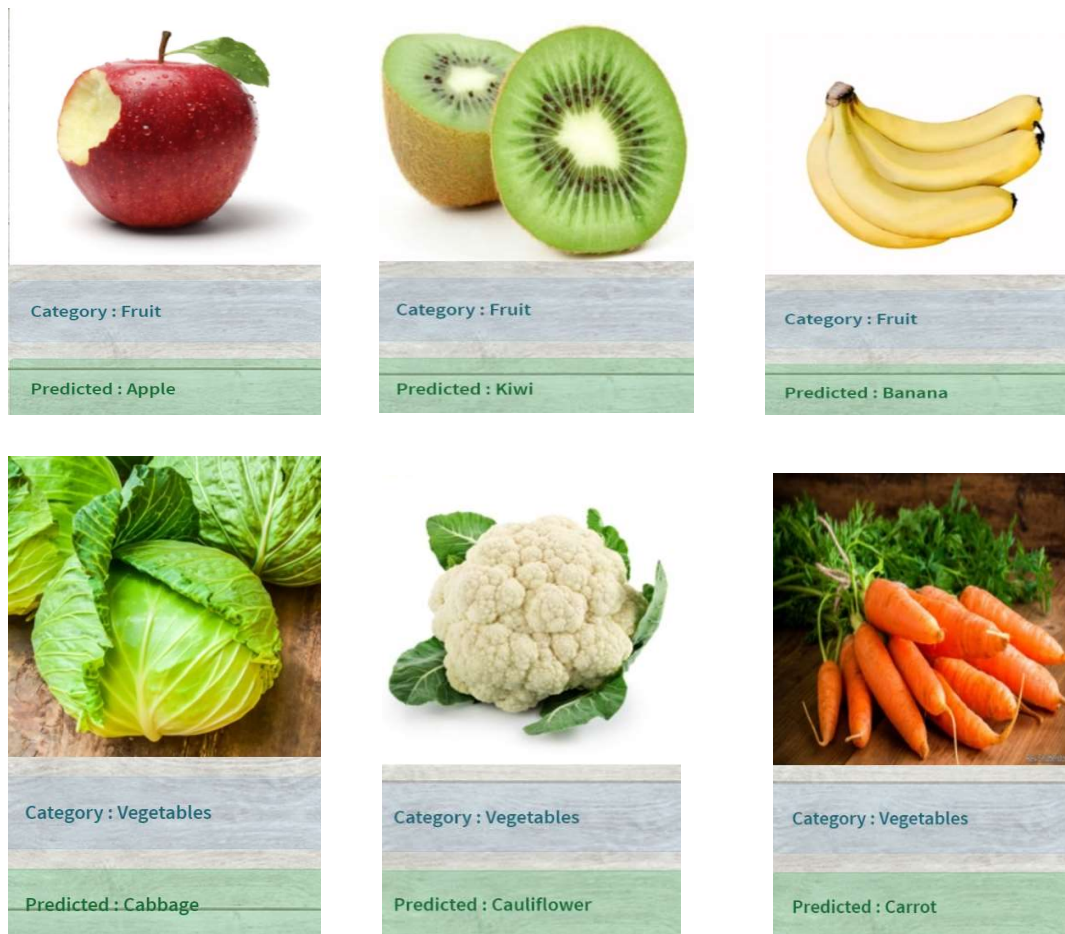


Fig.4.1: examples of model performance on different fruits and vegetables

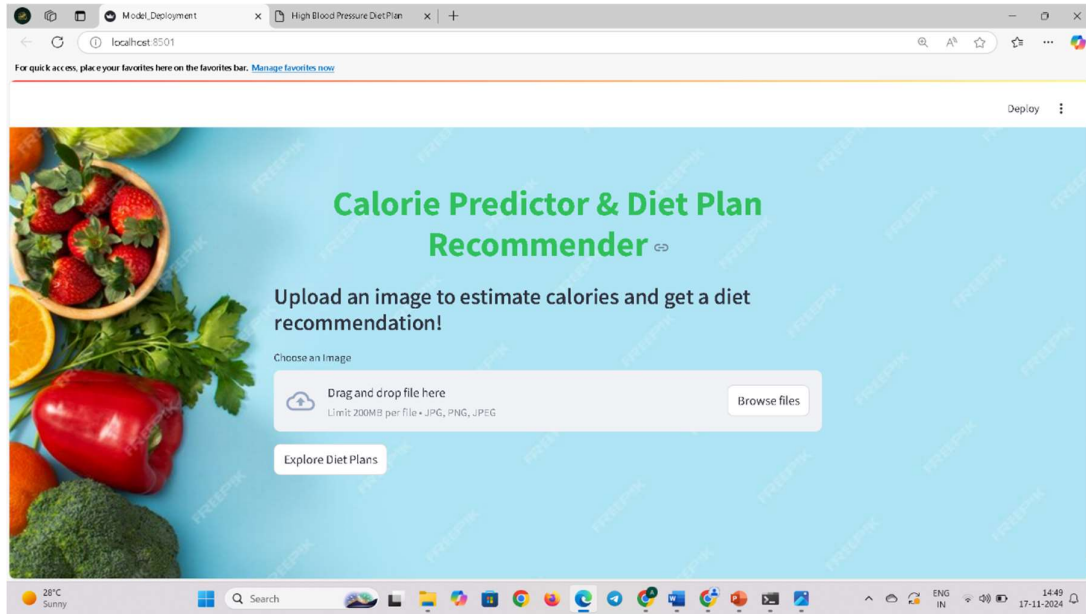


Fig.4.2: Calorie predictor and diet plan recommender main page

As discussed, the working of this web app is divided into two cases i.e., in the first case calorie and fat content is estimated in fruits and vegetable by uploading the image and it will predict the calorie and fat content present in input image and in the second case the dietary plan is suggested for the people who are suffering with some diseases like high blood pressure, rheumatoid arthritis, high cholesterol. Now, let us discuss each case

Case 1: Work Flow of Calorie and Fat content Extraction

1. User will upload the image. That image will be stored in local System.
2. Now we will resize the image according to our model shape, it will convert into vector
3. Now this vector will be passed to our model, our model will classify the class of category
4. We will get the ID of category, now we need to map the labels according to the ID.
5. Now our System will web-scrap the calories and total fat for the predicted class and the calories and total fat will be displayed as output in web application as shown in Fig.4.3 displays the calorie and fat content in cabbage and Fig 4.4 displays calorie and fat content in apple.

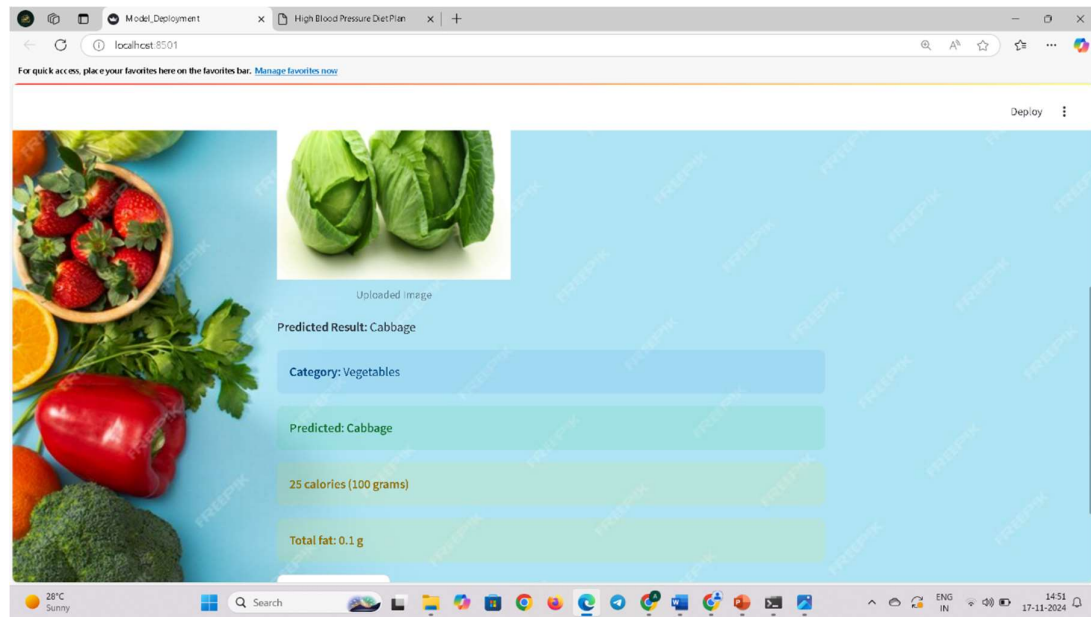


Fig.4.3 Calorie and fat count of cabbage

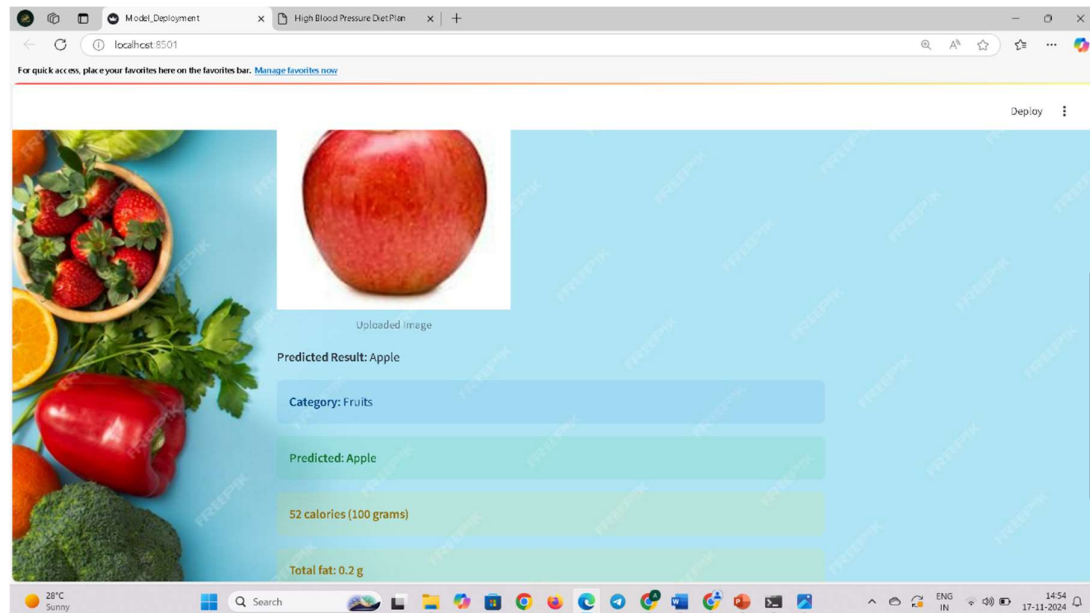


Fig.4.4: Calorie and fat count of apple

Case 2: Work Flow of weekly diet plans recommender

In the case 2 dietary plan is suggested for the people who are suffering with some diseases like high blood pressure, rheumatoid arthritis, high cholesterol. Here the user can view the information regarding diseases and the dietary plan recommended for

those diseases.

1. After clicking on that button user can able to view diseases like cardio, Obesity, Diabetes, rheumatoid arthritis, high cholesterol as shown in fig.4.5.
2. Each disease consists of two icons one is “know more” and the other is “view diet plan” as shown in Fig.4.6.
3. In the “Know more” option users can view the detailed information about the diseases like cause, symptoms, treatment methods and medication details as shown in fig.4.7.
4. In “view diet plan” the weekly diet plans are suggested for the people according to the diseases as mentioned above as shown in fig.4.8.

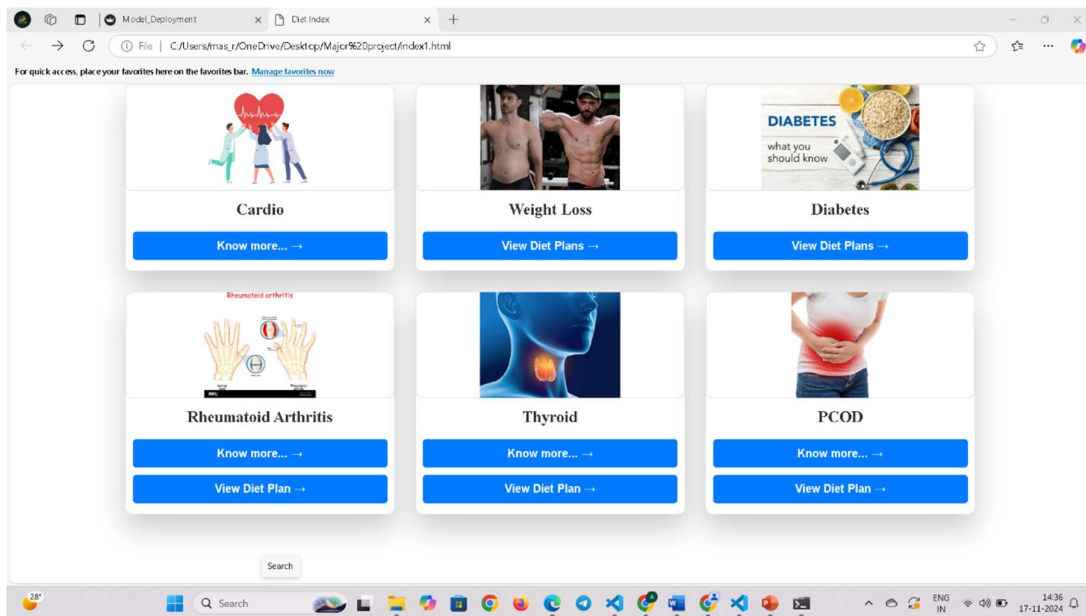


Fig.4.5: Main page of weekly diet plans recommender

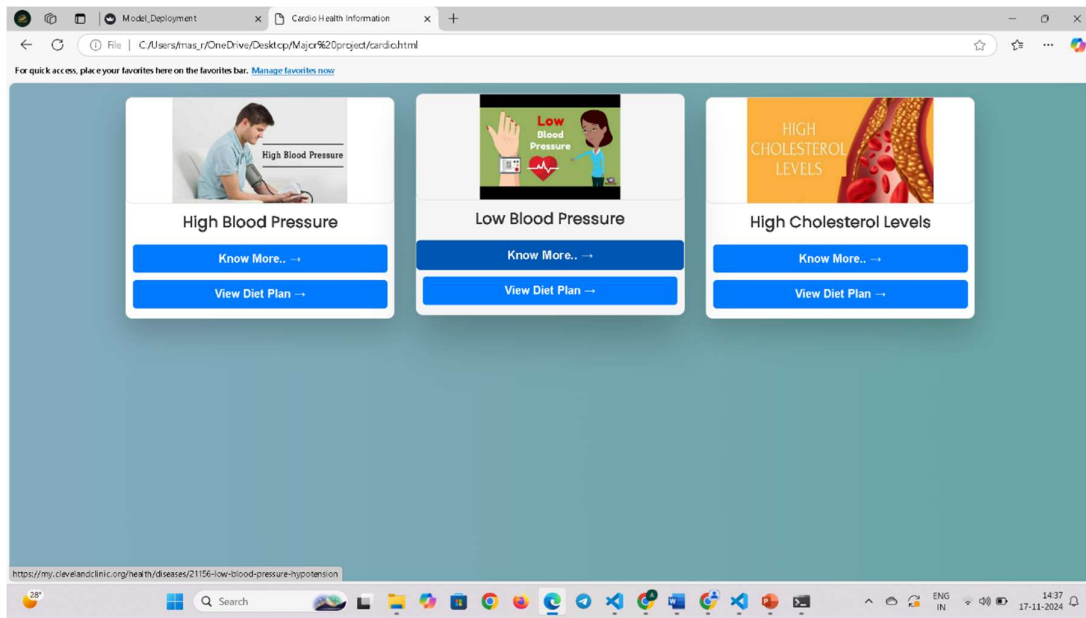


Fig.4.6: Diseases mentioned in cardio

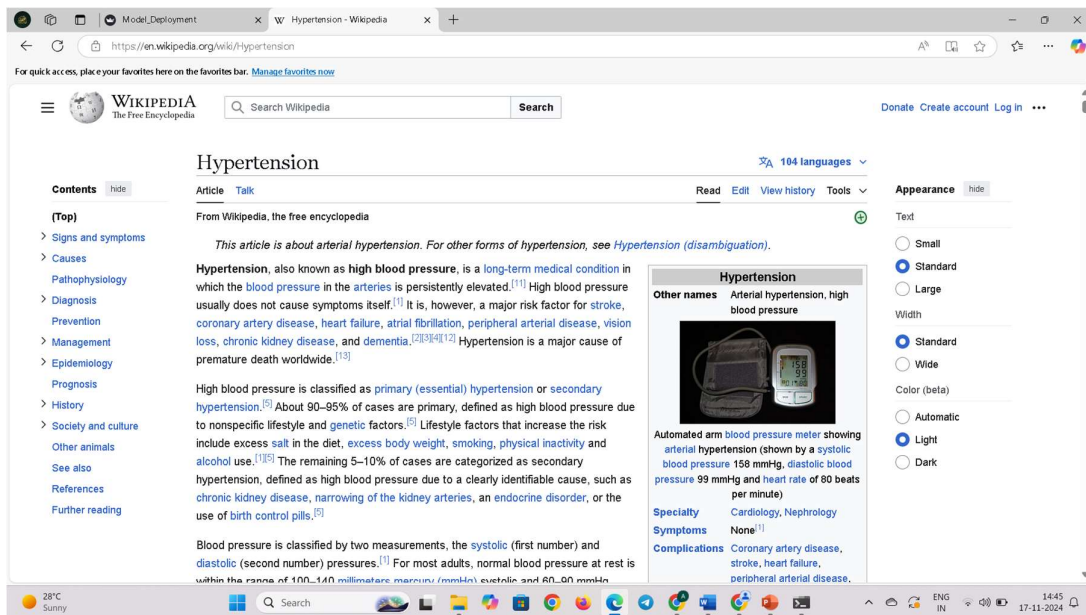


Fig.4.7: information regarding high blood pressure.

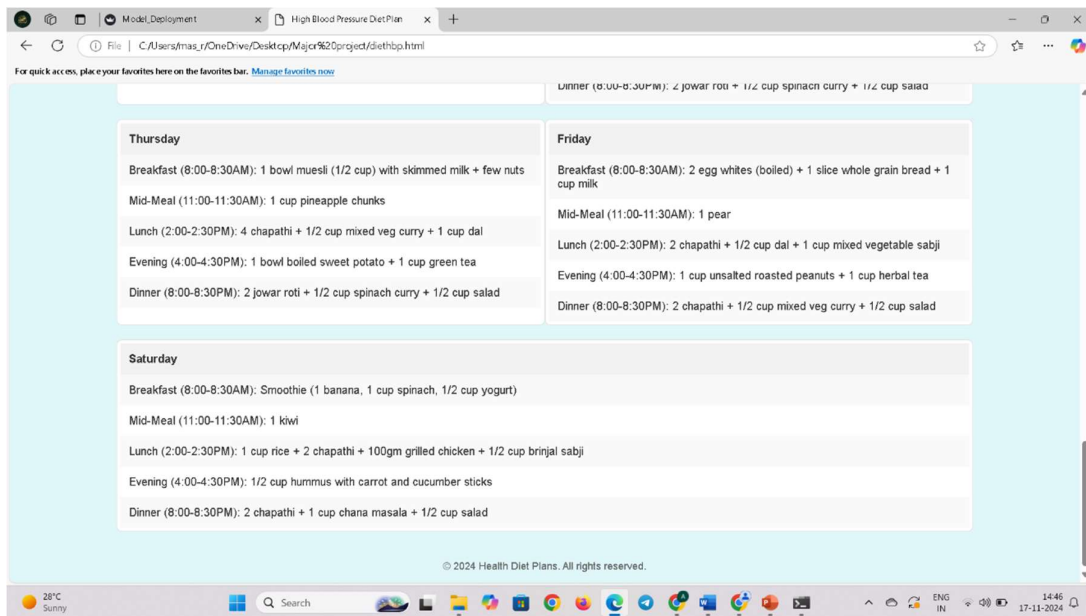
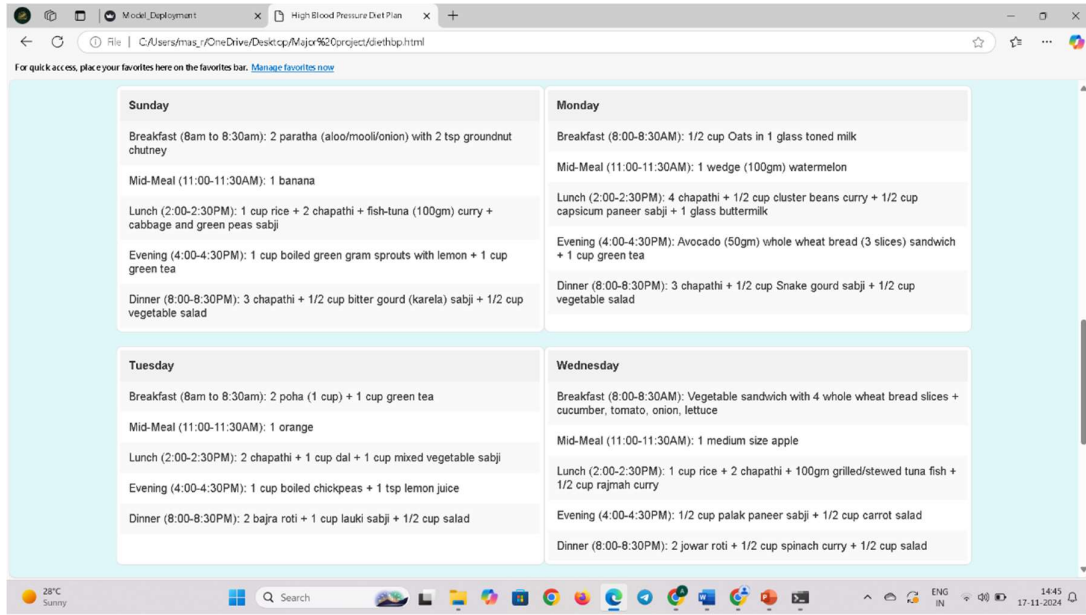


Fig.4.8: weekly dietary plan for the high blood pressure.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSIONS

Uncontrolled and imbalanced calorie intake are both main sources of problem that are causing overweight and obesity. However, these problems are not being taken seriously enough from the people. Aside from lack of awareness of the problem, one of the causes could be difficulty in acquiring good diet plans as some disease specific diets are not very well documented. Traditional ways of food logging is inefficient and discouraging as it requires a lot of effort to perform. Recent rise in popularity of deep learning had raised interest among many people in the field of machine learning. Not only it outperforms traditional machine learning techniques, it has a wide area of application in different fields. At the same time, rapid progression of technology development had made them much more affordable and increased capability to perform heavy computation locally.

This project proposed a web application that applied deep learning to acquire fruit and vegetable calorie and fat content to assist one to lead a healthy and longevity life by resolving dietary issues by providing weekly diet plans. Not only it provides an easier way to access information of food but could also serves as an educational tool for people to learn more about certain diseases. In a nutshell, the project aims to promote a healthy lifestyle by raising awareness among people towards the importance of balanced calorie intake. Hence, it can reduce the problem that the country is currently facing. Overall, the project has a potential to grow into a more usable tool as the classes of fruits and vegetables that it is able to identify increases.

5.2 FUTURE SCOPE

There are many improvements that could be added to enhance the project in many aspects. One of them could be implementation of object detection so that multiple fruits or vegetable objects can be detected and sum of calories can be computed instead. Further, non-vegetarian foods may also be categorized and provided with necessary protein contents as this is also one of necessary nutrient needed for a diet plan. Moreover, model performance can be further improved in terms of accuracy and fruit and vegetable variety. This would help to extend the usability of the project.

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Inventive Computing and Informatics (ICICI 2017) IEEE Xplore Compliant - Part Number: CFP17L34-ART, ISBN: 978-1-5386-4031-9, pp. 109-113.

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APPENDIX-I

CODE

Data_Training.ipynb

```
import numpy as np
import pandas as pd
import os
from pathlib import Path
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow.keras.preprocessing.image import load_img, img_to_array

print(tf.__version__)

# Define directories using raw strings to avoid escape character issues
train_dir = Path(r'C:\Users\mas_r\OneDrive\Desktop\Major project\Dataset\train')
train_filepaths = list(train_dir.glob(r'**/*.jpg'))

test_dir = Path(r'C:\Users\mas_r\OneDrive\Desktop\Major project\Dataset\test')
test_filepaths = list(test_dir.glob(r'**/*.jpg'))

val_dir = Path(r'C:\Users\mas_r\OneDrive\Desktop\Major project\Dataset\validation')
val_filepaths = list(val_dir.glob(r'**/*.jpg'))

def image_processing(filepath):
    """Create a DataFrame with the filepath and the labels of the pictures."""

    # Extract labels using Path to handle file paths correctly
    labels = [Path(fp).parts[-2] for fp in filepath]

    # Create DataFrame from filepaths and labels
    filepath_series = pd.Series(filepath, name='Filepath').astype(str)
    labels_series = pd.Series(labels, name='Label')
```

```

# Combine into a DataFrame
df = pd.concat([filepath_series, labels_series], axis=1)

# Shuffle the DataFrame
df = df.sample(frac=1).reset_index(drop=True)

return df

# Process the filepaths for train, test, and validation sets
train_df = image_processing(train_filepaths)
test_df = image_processing(test_filepaths)
val_df = image_processing(val_filepaths)

# Print some details about the training set
print('-- Training set --\n')
print(f'Number of pictures: {train_df.shape[0]}\n')
print(f'Number of different labels: {len(train_df.Label.unique())}\n')
print(f'Labels: {train_df.Label.unique()}')

df_unique = train_df.copy().drop_duplicates(subset=["Label"]).reset_index()
fig, axes = plt.subplots(nrows=6, ncols=6, figsize=(8, 7),
                        subplot_kw={'xticks': [], 'yticks': []})
for i, ax in enumerate(axes.flat):
    ax.imshow(plt.imread(df_unique.Filepath[i]))
    ax.set_title(df_unique.Label[i], fontsize=12)
plt.tight_layout(pad=0.5)
plt.show()

train_generator = tf.keras.preprocessing.image.ImageDataGenerator(
    preprocessing_function=tf.keras.applications.mobilenet_v2.preprocess_input
)
val_generator = tf.keras.preprocessing.image.ImageDataGenerator(
    preprocessing_function=tf.keras.applications.mobilenet_v2.preprocess_input

```

)

```
train_images = train_generator.flow_from_dataframe(  
    dataframe=train_df,  
    x_col='Filepath',  
    y_col='Label',  
    target_size=(224, 224),  
    color_mode='rgb',  
    class_mode='categorical',  
    batch_size=32,  
    shuffle=True,  
    seed=0,  
    rotation_range=30,  
    zoom_range=0.15,  
    width_shift_range=0.2,  
    height_shift_range=0.2,  
    shear_range=0.15,  
    horizontal_flip=True,  
    fill_mode="nearest"
```

)

```
val_images = val_generator.flow_from_dataframe(  
    dataframe=val_df,  
    x_col='Filepath',  
    y_col='Label',  
    target_size=(224, 224),  
    color_mode='rgb',  
    class_mode='categorical',  
    batch_size=32,  
    shuffle=True,  
    seed=0
```

)

```

# Define the model
pretrained_model = tf.keras.applications.MobileNetV2(
    input_shape=(224, 224, 3),
    include_top=False,
    weights='imagenet',
    pooling='avg'
)
pretrained_model.trainable = False

inputs = pretrained_model.input
x = tf.keras.layers.Dense(128, activation='relu')(pretrained_model.output)
x = tf.keras.layers.Dense(128, activation='relu')(x)
outputs = tf.keras.layers.Dense(len(train_df.Label.unique()), activation='softmax')(x)

model = tf.keras.Model(inputs=inputs, outputs=outputs)

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train the model
history = model.fit(
    train_images,
    validation_data=val_images,
    epochs=5,
    callbacks=[
        tf.keras.callbacks.EarlyStopping(
            monitor='val_loss',
            patience=2,
            restore_best_weights=True
        )
    ]
)

```

```

    ]
)

#Load test images
test_datagen = tf.keras.preprocessing.image.ImageDataGenerator(rescale=1./255)
test_images = test_datagen.flow_from_dataframe(
    dataframe=test_df,
    x_col='Filepath',
    y_col='Label',
    target_size=(224, 224),
    class_mode='categorical',
    shuffle=False
)

# Load test images with the same preprocessing as training images
test_datagen = tf.keras.preprocessing.image.ImageDataGenerator(
    preprocessing_function=tf.keras.applications.mobilenet_v2.preprocess_input
)

test_images = test_datagen.flow_from_dataframe(
    dataframe=test_df,
    x_col='Filepath',
    y_col='Label',
    target_size=(224, 224),
    class_mode='categorical',
    shuffle=False
)

# Predict the labels of the test_images
pred = model.predict(test_images)
pred = np.argmax(pred, axis=1)

# Map the labels

```

```

labels = (train_images.class_indices)
labels = dict((v, k) for k, v in labels.items())
pred1 = [labels[k] for k in pred]

def output(location):
    img = load_img(location, target_size=(224, 224))
    img = img_to_array(img)
    img = img / 255.0
    img = np.expand_dims(img, axis=0)
    answer = model.predict(img)
    y_class = answer.argmax(axis=-1)
    res = labels[y_class[0]] # Get the label
    return res

# Test the output function
img_result = output(r"C:\Users\mas_r\OneDrive\Desktop\Major
project\Dataset\test\cabbage\Image_1.jpg")
print(img_result)

# Save the model
model.save('FV.h5')

```

Model Deployment:

Model_Deployment.py

```
import streamlit as st
from PIL import Image
from keras.preprocessing.image import load_img, img_to_array
import numpy as np
from keras.models import load_model
import requests
from bs4 import BeautifulSoup
import webbrowser
import os
import base64

# Load the model
model = load_model('FV.h5')

# Mapping of class indices to labels
labels = {
    0: 'apple', 1: 'banana', 2: 'beetroot', 3: 'capsicum', 4: 'cabbage', 5: 'capsicum', 6: 'carrot',
    7: 'cauliflower',
    8: 'chilli pepper', 9: 'corn', 10: 'cucumber', 11: 'eggplant', 12: 'garlic', 13: 'ginger', 14:
    'grapes',
    15: 'jalapeno', 16: 'kiwi', 17: 'lemon', 18: 'lettuce', 19: 'mango', 20: 'onion', 21: 'orange',
    22: 'paprika',
    23: 'pear', 24: 'peas', 25: 'pineapple', 26: 'pomegranate', 27: 'potato', 28: 'raddish', 29:
    'soy beans',
    30: 'spinach', 31: 'sweetcorn', 32: 'sweetpotato', 33: 'tomato', 34: 'turnip', 35:
    'watermelon'
}

fruits = ['Apple', 'Banana', 'Chilli Pepper', 'Grapes', 'Jalapeno', 'Kiwi', 'Lemon', 'Mango',
'Orange', 'Paprika', 'Pear', 'Pineapple', 'Pomegranate', 'Watermelon']
```



```
vegetables = ['Beetroot', 'Cabbage', 'Capsicum', 'Carrot', 'Cauliflower', 'Corn', 'Cucumber',  
'Eggplant', 'Ginger', 'Lettuce', 'Onion', 'Peas', 'Potato', 'Raddish', 'Soy Beans', 'Spinach',  
'Sweetcorn', 'Sweetpotato', 'Tomato', 'Turnip']
```

```
def fetch_calories(prediction):
```

```
    try:
```

```
        url = 'https://www.google.com/search?q=calories in ' + prediction + ' 100g'
```

```
        headers = {'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64)
```

```
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/58.0.3029.110 Safari/537.3'}
```

```
        req = requests.get(url, headers=headers)
```

```
        if req.status_code != 200:
```

```
            return None
```

```
        soup = BeautifulSoup(req.text, 'html.parser')
```

```
        calories_info = soup.find_all("div", class_="BNeawe iBp4i AP7Wnd")
```

```
        if calories_info:
```

```
            for info in calories_info:
```

```
                text = info.get_text()
```

```
                if "calories" in text.lower():
```

```
                    return text
```

```
        return None
```

```
    except Exception as e:
```

```
        return None
```

```
def fetch_fat(prediction):
```

```
    try:
```

```
        url = 'https://www.google.com/search?q=fat in ' + prediction
```

```
        headers = {'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64)
```

```
AppleWebKit/537.36 (KHTML, like Gecko) Chrome/58.0.3029.110 Safari/537.3'}
```

```
        req = requests.get(url, headers=headers).text
```

```
        scrap = BeautifulSoup(req, 'html.parser')
```

```
        fat_info = scrap.find("div", class_="BNeawe iBp4i AP7Wnd")
```

```
        if fat_info:
```

```

        return fat_info.text
    else:
        return None
except Exception as e:
    return None

def get_base64_of_bin_file(bin_file):
    with open(bin_file, 'rb') as f:
        data = f.read()
    return base64.b64encode(data).decode()

def set_bg_hack(main_bg):
    main_bg_ext = "png"
    st.markdown(
        f"""
        <style>
        .stApp {{
            background:
url(data:image/{main_bg_ext};base64,{base64.b64encode(open(main_bg,
"rb").read()).decode()});
            background-size: cover;
        }}
        </style>
        """,
        unsafe_allow_html=True
    )

def processed_img(img_path):
    img = load_img(img_path, target_size=(224, 224))
    img = img_to_array(img)
    img = img / 255
    img = np.expand_dims(img, [0])
    answer = model.predict(img)

```

```

y_class = answer.argmax(axis=-1)
y = int(y_class[0])
res = labels[y]
return res.capitalize()

def run():
    # Set the background image (replace with your own image path)
    bg_image_path = r"C:\\Users\\mas_r\\OneDrive\\Desktop\\Major
project\\Images\\bg1.jpg"
    if os.path.exists(bg_image_path):
        set_bg_hack(bg_image_path)

    st.markdown("<h1 style='text-align: center; color: #31c05c;'>Calorie Predictor & Diet
Plan Recommender</h1>", unsafe_allow_html=True)
    st.write("### Upload an image to estimate calories and get a diet recommendation!")

    # Image upload
    img_file = st.file_uploader("Choose an Image", type=["jpg", "png"])

    if img_file is not None:
        img = Image.open(img_file).resize((250, 250))
        st.image(img, caption="Uploaded Image", use_column_width=False, width=300)

        # Save uploaded image
        upload_dir = './upload_images'
        if not os.path.exists(upload_dir):
            os.makedirs(upload_dir)

        save_image_path = os.path.join(upload_dir, img_file.name)
        with open(save_image_path, "wb") as f:
            f.write(img_file.getbuffer())

        # Predict the category (fruit or vegetable) and fetch related information

```

```

try:
    result = processed_img(save_image_path)
    st.write(f"**Predicted Result:** {result}")

    if result in vegetables:
        st.info(f"**Category:** Vegetables'")
    else:
        st.info(f"**Category:** Fruits'")

    st.success(f"**Predicted: {result}**")

    cal = fetch_calories(result)
    fat = fetch_fat(result)

    if cal:
        st.warning(f"**{cal} (100 grams)**")
    if fat:
        st.warning(f"**Total fat: {fat}**")

except Exception as e:
    st.error(f"Error during image processing: {e}")

# Button to navigate to diet plans
if st.button('Explore Diet Plans', help="Get personalized diet plans"):
    webbrowser.open_new_tab(r"C:\\Users\\mas_r\\OneDrive\\Desktop\\Major
project\\index1.html")

if __name__ == "__main__":
    run()

```

Web Templates for diet recommendation:

index1.html

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Diet Index</title>
    <link rel="stylesheet" href="mystyle.css" />
  </head>

  <body class="diet-background">
    <div class="grid">
      <div class="grid-item">
        <div class="card">
          
          <div class="card-content">
            <h1 class="card-header">Cardio</h1>
            <a href="cardio.html">
              <button class="card-btn">Know more... <span>&rarr;</span></button>
            </a>
          </div>
        </div>
      </div>

      <div class="grid-item">
        <div class="card">
          
          <div class="card-content">
            <h1 class="card-header">Weight Loss</h1>
            <a href="dietwl.html">
              <button class="card-btn">View Diet Plans <span>&rarr;</span></button>
            </a>
          </div>
        </div>
      </div>
    </div>
  </body>
</html>
```

```

        </a>
    </div>
</div>

<div class="grid-item">
    <div class="card">
        
        <div class="card-content">
            <h1 class="card-header">Diabetes</h1>
            <a href="dietdbt.html">
                <button class="card-btn">View Diet Plans <span>&rarr;</span></button>
            </a>
        </div>
    </div>
</div>

<div class="grid-item">
    <div class="card">
        
        <div class="card-content">
            <h1 class="card-header">Rheumatoid Arthritis</h1>
            <a href="https://www.niams.nih.gov/health-topics/rheumatoid-arthritis">
                <button class="card-btn">Know more... <span>&rarr;</span></button>
            </a><br>
            <a href="dietra.html">
                <button class="card-btn">View Diet Plan <span>&rarr;</span></button>
            </a>
        </div>
    </div>
</div>

<div class="grid-item">

```

```

<div class="card">
  
  <div class="card-content">
    <h1 class="card-header">Thyroid</h1>
    <a href="https://my.clevelandclinic.org/health/diseases/8541-thyroid-disease">
      <button class="card-btn">Know more... <span>&rarr;</span></button>
    </a><br>
    <a href="dietthyroid.html">
      <button class="card-btn">View Diet Plan <span>&rarr;</span></button>
    </a>
  </div>
</div>
</div>
</div>

<div class="grid-item">
  <div class="card">
    
    <div class="card-content">
      <h1 class="card-header">PCOD</h1>
      <a href="https://www.maxhealthcare.in/blogs/are-you-suffering-from-
pcod#:~:text=Polycystic%20Ovarian%20Disease%20(PCOD)%20is,in%20the%20ovarie
s%20over%20time.">
        <button class="card-btn">Know more... <span>&rarr;</span></button>
      </a><br>
      <a href="dietpcod.html">
        <button class="card-btn">View Diet Plan <span>&rarr;</span></button>
      </a>
    </div>
  </div>
</div>
</div>
</div>
</body>
</html>

```

mystyle.css

```
@import url("https://fonts.googleapis.com/css?family=Poppins&display=swap");
```

```
* {  
  margin: 0;  
  padding: 0;  
}  
html {  
  box-sizing: border-box;  
  font-size: 62.5%;  
}  
body {  
  background-color: #eee;  
  font-family: "Poppins", sans-serif;  
  display: flex;  
  justify-content: center;  
  align-items: center;  
  min-height: 100vh;  
}  
.grid {  
  display: grid;  
  width: 114em;  
  grid-gap: 6rem;  
  grid-template-columns: repeat(auto-fit, minmax(30rem, 1fr));  
  align-items: start;  
}  
.grid-item {  
  background-color: #fff;  
  border-radius: 0.4rem;  
  overflow: hidden;  
  box-shadow: 0 3rem 6rem rgba(0, 0, 0, 0.1);  
  cursor: pointer;  
  transition: 0.2s;
```



```
}  
.grid-item:hover {  
  transform: translateY(-0.5%);  
  box-shadow: 0 4rem 8rem rgba(0, 0, 0, 0.5);  
}  
.card-img {  
  display: block;  
  width: 100%;  
  height: 20rem;  
  object-fit: cover;  
}  
.card-content {  
  padding: 3rem;  
}  
.card-header {  
  font-size: 3rem;  
  font-weight: 500;  
  color: #0d0d0d;  
  margin-bottom: 1.5rem;  
}  
.card-text {  
  font-size: 1.6rem;  
  letter-spacing: 0.1rem;  
  line-height: 1.7;  
  color: #3d3d3d;  
  margin-bottom: 2.5rem;  
}  
.card-btn {  
  display: block;  
  width: 100%;  
  padding: 1.5rem;  
  font-size: 2rem;  
  text-align: center;
```

```

    color: #3363ff;
    background-color: #d8e0fd;
    border: none;
    border-radius: 0.4rem;
    transition: 0.2s;
    cursor: pointer;
    letter-spacing: 0.1rem;
}
.card-btn span {
    margin-left: 1rem;
    transition: 0.2s;
}
.card-btn:hover,
.card-btn:active {
    background-color: #c2cffe;
}
.card-btn:hover span,
.card-btn:active span {
    margin-left: 1.5rem;
}
@media only screen and (max-width: 60em) {
    body {
        padding: 3rem;
    }
    .grid {
        grid-gap: 3rem;
    }
}

```

Cardio.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />

```

```

<meta name="viewport" content="width=device-width, initial-scale=1.0" />
<title>Cardio</title>

<link rel="stylesheet" href="mystyle.css" />
</head>

<body>
<div class="grid">
  <div class="grid-item">
    <div class="card">
      
      <div class="card-content">
        <h1 class="card-header">High blood pressure</h1>

        <a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/hbp.html">
          <button class="card-btn">Know more.. <span>&rarr;</span></button>

        </a><br>

        <a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/dietHBP.ht
ml">
          <button class="card-btn">View diet plan <span>&rarr;</span></button>

        </a>

      </div>
    </div>
  </div>
</div>
<div class="grid-item">

```

```

<div class="card">
  
  <div class="card-content">
    <h1 class="card-header">Low Blood Pressure</h1>
    <a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/lbp.html">
      <button class="card-btn">Know more.. <span>&rarr;</span></button>

    </a><br>

    <a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/dietlbp.html
">
      <button          class="card-btn">view          diet          plan
<span>&rarr;</span></button></a>
    </div>
  </div>
</div>
<div class="grid-item">
  <div class="card">
    
    <div class="card-content">
      <h1 class="card-header">High Cholesterol levels</h1>
      <a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/cl.html">
        <button class="card-btn">Know more.. <span>&rarr;</span></button>

```

```
</a><br>
<a
href="file:///C:/Users/Geetha/OneDrive/Desktop/project/Fruit_Vegetable_Calorie
s_Prediction/Fruit_Vegetable_Calories_Prediction/mydiet%20plan/dietCL.html
">
<button class="card-btn">view diet plan <span>&rarr;</span></button></a>
</div>
</div>
</div>
</div>
</body>
</html>
```

Style.css

```
*
{
    padding:0;
    margin:0;
}
.container
{padding:30px;
    font-family:'Roboto';
    font-size:16px;
    font-weight:300;
    line-height:24px;
}
.container p
{
    margin-bottom:15px;
}
thumbnail
{
    width:450px;
    position:relative;
    box-shadow:0 0 5px rgba(0,0,0,.4);
    background-color:#fff;
    overflow:hidden;
    border-radius:8px;
    margin-bottom:10px;
}
.thumbnail:before
{
    content:"";
    position:absolute;
    top:0;
    left:0;
```

```

        width:100%;
        height:100%;
        border:8px solid #fff;
        z-index:2;
        overflow:hidden;
        box-sizing:border-box;
    }
    .thumbanail img
    {
        width:100%;
        height:auto;
        border-radius:8px;
        transform:scale(1) rotate(0deg);
        transition:all .5s linear;
    }
    .thumbanail:hover img
    {
        transform:scale(1.1) rotate(-5deg);
    }

    .thumbanail.right
    {
        float:right;
        margin-left:20px;
    }
    .thumbanail.left
    {
        float:left;
        margin-right:20px;
    }

    @media screen and (max-width:767px)

```

```
{  
  
    .thumbanail.right  
    {  
        float:none;  
        margin-left:0px;  
    }  
    .thumbanail.left  
    {  
        float:none;  
        margin-right:0px;  
    }  
    .thumbanail{  
        width:100%;  
    }  
}
```


Weekly diet plan code:

dietHBP.html

```
<html>
<head>
  <title>High Blood Pressure Diet Plan</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #e0f7fa; /* Light blue background color */
      color: #333; /* Darker text color for readability */
      margin: 0;
      padding: 0;
    }

    h1 {
      font-size: 30px;
      color: white;
      background-color: blue;
      text-align: center;
      padding: 15px;
    }

    .container {
      width: 80%;
      margin: 0 auto;
    }

    .image-container {
      text-align: center;
      margin: 20px 0;
    }
```

```
.image-container img {
    width: 70%; /* Adjusted image width for better appearance */
    max-width: 600px; /* Max width to prevent enlargement */
    height: auto; /* Maintain aspect ratio */
    border-radius: 8px; /* Rounded corners for the image */
    box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1); /* Subtle shadow for depth */
}
```

```
.summary {
    background-color: #e6f7ff; /* Light background for summary */
    border: 1px solid #b3e0ff; /* Border for summary */
    padding: 15px;
    margin-bottom: 20px;
    border-radius: 8px;
}
```

```
.row {
    display: flex;
    flex-wrap: wrap;
    justify-content: space-between;
}
```

```
.column {
    flex: 48%; /* Adjusted for spacing */
    padding: 5px;
    margin-bottom: 20px;
    border: 1px solid #ddd; /* Border around columns */
    border-radius: 8px; /* Rounded corners */
    background-color: white; /* White background for columns */
}
```

```
table {
    border-collapse: collapse;
```

```

        width: 100%;
    }

    th, td {
        text-align: left;
        padding: 12px;
    }

    th {
        background-color: #f2f2f2; /* Header background color */
    }

    tr:nth-child(even) {
        background-color: #f9f9f9; /* Zebra stripes for tables */
    }

    footer {
        text-align: center;
        margin-top: 20px;
        font-size: 14px;
        color: #666;
    }
</style>
</head>
<body>
    <div class="container">
        <h1>HIGH BLOOD PRESSURE WEEKLY DIET PLAN</h1>
        <div class="image-container">
            
        </div>

        <div class="summary">
            <h3>Weekly Summary</h3>

```

<p>This diet plan is designed to help manage high blood pressure effectively. It includes a variety of healthy food items that are low in sodium and rich in nutrients. Follow this plan to maintain a balanced diet while keeping your blood pressure in check.</p>

</div>

<h3>Do's And Don'ts While Following Diet Plan</h3>

<p>Add these food items and habits in your lifestyle if you are following high blood pressure diet plan:</p>

Eat less salt

Include more fruits and vegetables

Include whole grain cereals

Maintain a healthy weight

Include lean meat, fish, and poultry

Include low-fat dairy products

Try to avoid these habits if you are following high blood pressure diet plan:

Alcohol

Smoking

Sedentary lifestyle

Sleep deprivation

<h3>Food Items You Can Easily Consume In High Blood Pressure Diet Plan</h3>

<p>Banana: Portable, low in sodium, and rich in potassium.

Spinach: Low in calories and high in potassium, folate, and magnesium.

Celery: Contains phthalides that relax artery walls.

Oatmeal: High in fiber and helps maintain healthy blood pressure.

Avocado: Reduces blood pressure and cholesterol levels.

Watermelon: Contains L-citrulline, known to lower blood pressure.

Beetroot: Rich in nitrates, improves blood flow.

Oranges: High in vitamin C, supports blood pressure health.

<div class="row">

<div class="column">

<table>

<tr>

<th>Sunday</th>

</tr>

<tr>

<td>Breakfast (8am to 8:30am): 2 paratha (aloo/mooli/onion) with 2 tsp groundnut chutney</td>

</tr>

<tr>

<td>Mid-Meal (11:00-11:30AM): 1 banana</td>

</tr>

<tr>

<td>Lunch (2:00-2:30PM): 1 cup rice + 2 chapathi + fish-tuna (100gm) curry + cabbage and green peas sabji</td>

</tr>

<tr>

<td>Evening (4:00-4:30PM): 1 cup boiled green gram sprouts with lemon + 1 cup green tea</td>

</tr>

<tr>

<td>Dinner (8:00-8:30PM): 3 chapathi + 1/2 cup bitter gourd (karela) sabji + 1/2 cup vegetable salad</td>

</tr>

```

    </table>
  </div>
  <div class="column">
    <table>
      <tr>
        <th>Monday</th>
      </tr>
      <tr>
        <td>Breakfast (8:00-8:30AM): 1/2 cup Oats in 1 glass toned
milk</td>
      </tr>
      <tr>
        <td>Mid-Meal (11:00-11:30AM): 1 wedge (100gm) watermelon</td>
      </tr>
      <tr>
        <td>Lunch (2:00-2:30PM): 4 chapathi + 1/2 cup cluster beans curry
+ 1/2 cup capsicum paneer sabji + 1 glass buttermilk</td>
      </tr>
      <tr>
        <td>Evening (4:00-4:30PM): Avocado (50gm) whole wheat bread
(3 slices) sandwich + 1 cup green tea</td>
      </tr>
      <tr>
        <td>Dinner (8:00-8:30PM): 3 chapathi + 1/2 cup Snake gourd sabji
+ 1/2 cup vegetable salad</td>
      </tr>
    </table>
  </div>
</div>

<div class="row">
  <div class="column">
    <table>

```

```

<tr>
  <th>Tuesday</th>
</tr>
<tr>
  <td>Breakfast (8am to 8:30am): 2 poha (1 cup) + 1 cup green tea</td>
</tr>
<tr>
  <td>Mid-Meal (11:00-11:30AM): 1 orange</td>
</tr>
<tr>
  <td>Lunch (2:00-2:30PM): 2 chapathi + 1 cup dal + 1 cup mixed
vegetable sabji</td>
</tr>
<tr>
  <td>Evening (4:00-4:30PM): 1 cup boiled chickpeas + 1 tsp lemon
juice</td>
</tr>
<tr>
  <td>Dinner (8:00-8:30PM): 2 bajra roti + 1 cup lauki sabji + 1/2 cup
salad</td>
</tr>
</table>
</div>
<div class="column">
  <table>
    <tr>
      <th>Wednesday</th>
    </tr>
    <tr>
      <td>Breakfast (8:00-8:30AM): Vegetable sandwich with 4 whole
wheat bread slices + cucumber, tomato, onion, lettuce</td>
    </tr>
    <tr>

```

<td>Mid-Meal (11:00-11:30AM): 1 medium size apple</td>
</tr>
<td>Lunch (2:00-2:30PM): 1 cup rice + 2 chapathi + 100gm grilled/stewed tuna fish + 1/2 cup rajmah curry</td>
</tr>
<td>Evening (4:00-4:30PM): 1/2 cup palak paneer sabji + 1/2 cup carrot salad</td>
</tr>
<td>Dinner (8:00-8:30PM): 2 jowar roti + 1/2 cup spinach curry + 1/2 cup salad</td>
</tr>

Thursday
Breakfast (8:00-8:30AM): 1 bowl muesli (1/2 cup) with skimmed milk + few nuts
Mid-Meal (11:00-11:30AM): 1 cup pineapple chunks

<td>Lunch (2:00-2:30PM): 4 chapathi + 1/2 cup mixed veg curry +
1 cup dal</td>

</tr>

<tr>

<td>Evening (4:00-4:30PM): 1 bowl boiled sweet potato + 1 cup
green tea</td>

</tr>

<tr>

<td>Dinner (8:00-8:30PM): 2 jowar roti + 1/2 cup spinach curry +
1/2 cup salad</td>

</tr>

</table>

</div>

<div class="column">

<table>

<tr>

<th>Friday</th>

</tr>

<tr>

<td>Breakfast (8:00-8:30AM): 2 egg whites (boiled) + 1 slice whole
grain bread + 1 cup milk</td>

</tr>

<tr>

<td>Mid-Meal (11:00-11:30AM): 1 pear</td>

</tr>

<tr>

<td>Lunch (2:00-2:30PM): 2 chapathi + 1/2 cup dal + 1 cup mixed
vegetable sabji</td>

</tr>

<tr>

<td>Evening (4:00-4:30PM): 1 cup unsalted roasted peanuts + 1 cup
herbal tea</td>

</tr>

```

        <tr>
            <td>Dinner (8:00-8:30PM): 2 chapathi + 1/2 cup mixed veg curry +
1/2 cup salad</td>
        </tr>
    </table>
</div>
</div>

```

```

<div class="row">
    <div class="column">
        <table>
            <tr>
                <th>Saturday</th>
            </tr>
            <tr>
                <td>Breakfast (8:00-8:30AM): Smoothie (1 banana, 1 cup spinach,
1/2 cup yogurt)</td>
            </tr>
            <tr>
                <td>Mid-Meal (11:00-11:30AM): 1 kiwi</td>
            </tr>
            <tr>
                <td>Lunch (2:00-2:30PM): 1 cup rice + 2 chapathi + 100gm grilled
chicken + 1/2 cup brinjal sabji</td>
            </tr>
            <tr>
                <td>Evening (4:00-4:30PM): 1/2 cup hummus with carrot and
cucumber sticks</td>
            </tr>
            <tr>
                <td>Dinner (8:00-8:30PM): 2 chapathi + 1 cup chana masala + 1/2
cup salad</td>
            </tr>
        </table>
    </div>
</div>

```

```
        </table>
    </div>
</div>

<footer>
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</footer>
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